This is the authors' version of the following article: Abdi Tabari, M., Lee, H., & Farahanynia, M. (2026). Impact of methodological differences in task sequencing on second language learning: A multi-level meta-analysis. *International Review of Applied Linguistics in Language Teaching, 64*(2), 819–867. The version of record is available at https://doi.org/10.1515/iral-2024-0219. This version differs from the published article in layout and pagination; please cite the published version.

Impact of methodological differences in task sequencing on second language learning: a multi-level meta-analysis

Mahmoud Abdi Tabari*, Hansol Lee and Mahsa Farahanynia

Mahmoud Abdi Tabari, Department of English, University of Nevada, Frandsen Humanities, Room 105, Reno, NV 89557, USA. https://orcid.org/0000-0002-8022-5415

Hansol Lee, Department of English, Korea Military Academy, Seoul, Korea. https://orcid.org/0000-0002-6912-7128

Mahsa Farahanynia, Department of English Language, Pishtazan Higher Education Institute, Shiraz, Iran. https://orcid.org/0000-0003-3038-1871

***Corresponding author: Mahmoud Abdi Tabari,** Department of English, University of Nevada, Frandsen Humanities, Room 105, Reno, NV 89557, USA, E-mail: Mabditabari@unr.edu. https://orcid.org/0000-0002-8022-5415

Abstract

Drawing on the SSARC model, TBLT research has investigated the role of task sequencing in L2 production, yielding mixed results. This study conducts a multi-level meta-analysis of methodological practices in task sequencing to provide more transparency. Sixteen studies published up to May 2024 were selected based on inclusion criteria, and each study was coded for analyzed features and contextual variables. Methodological aspects, including study designs, sampling procedures, analyses, and reporting practices, were also coded. Significant methodological variations were identified across studies, including differences in L2 proficiency, learning contexts, task modalities, types, complexities, delivery methods, and constructs. Task complexity emerged as a significant factor influencing L2 performance, with more complex tasks yielding larger effect sizes. Additionally, simple and complex sequencing tasks exhibited a medium effect size in enhancing accuracy, particularly in between-group studies. These findings highlight the importance of balanced learner representation and standardized research practices to improve L2 learning outcomes.

Keywords: the SSARC model; methodological choices; moderating variables; a multi-level meta-analysis

1 Introduction

The examination of methodological differences in task sequencing (TS) within the field of second language (L2) acquisition remains a critical area of inquiry for both researchers and practitioners in Task-Based Language Teaching (TBLT). While TS, defined as the strategic arrangement of instructional tasks based on cognitive factors and sequenced from simple to complex to optimize learning outcomes, has been recognized as a cornerstone of effective L2 pedagogy (Abdi Tabari et al. 2023; Robinson 2010), recent scholarship highlights the need to address criticisms and uncertainties surrounding this construct. For instance, Skehan (2016) has questioned the theoretical underpinnings and practical applications of task sequencing, citing inconsistencies in findings related to task characteristics, resource-directing variables, and the inductive versus deductive approaches to sequencing. Similarly, Ellis et al. (2020) summarized skepticism toward the field, pointing to the inadequacy of theory-driven predictions, such as those proposed by the Cognition Hypothesis (CH) and the SSARC model, to consistently explain empirical outcomes.

Despite these critiques, the potential of TS to refine pedagogical strategies, inform syllabus design, and enhance the effectiveness of language education programs remains significant (Baralt et al. 2014). Addressing these challenges head-on, this study critically engages with the issues raised by Skehan (2016) and Ellis et al. (2020), incorporating their concerns into the analysis to provide a nuanced understanding of TS and its implications. By systematically analyzing methodological differences and their impact on learning outcomes, this meta-analysis seeks to clarify the theoretical and empirical foundations of TS, offering insights that can guide future research and practice.

Over the past two decades, a growing body of research has examined different orders of task sequencing – simple, medium, and complex (SC); complex, medium, and simple (CS); randomized (RAN); or isolated task performance (No sequence) – and their influence on L2 learning. This research is largely framed by Robinson's (2010) SSARC model, which emphasizes the importance of sequencing tasks to stabilize, simplify, automatize, reconstruct, and complexify cognitive processes. In this context, "*simple*" tasks are those that are designed to stabilize the learner's existing knowledge by reducing cognitive load and focusing on basic skills or structures that are already automatized, such as routine sentence construction or simple vocabulary recall. "*Medium*" tasks are aimed at simplifying learning by requiring the learner to process more complex linguistic forms or engage in tasks that push learners slightly beyond their current abilities but are still manageable, such as constructing sentences in varied contexts or engaging in simple conversations. "*Complex*" tasks are designed to challenge the learner by requiring more advanced cognitive processing, such as producing extended

discourse, engaging in problem-solving tasks, or using less familiar linguistic structures, thus encouraging the learner to move beyond their current competence to reconstruct and complexify their language skills. These different levels of task complexity are used to explore how task sequencing influences outcomes such as production (syntactic complexity, accuracy, lexical complexity, and fluency, or CALF), acquisition (the development of specific linguistic forms), and interaction (Malicka 2018).

These studies have underscored the importance of Robinson's (2010) proposal of task sequencing in TBLT, with scholars employing a wide array of methodological approaches. However, the resulting landscape of findings is marked by inconsistency and, at times, contradiction. While some studies report significant advantages of certain sequencing orders (especially the simple, medium, and complex order), others find minimal or no effects (see a systematic review by Jin and Yan 2024). These inconsistencies likely stem from variations in study design, participant characteristics, instructional contexts, and measurement tools (Ellis et al. 2020). However, there is no empirical evidence to corroborate the reasons possibly leading to the inconsistencies observed in previous research findings.

Given these divergent findings, there is a need for a rigorous analysis of existing research to uncover the underlying reasons for the impact(s) of task sequencing on L2 learning outcomes (Kim 2023). Equally important, identifying and categorizing methodological variations across prior studies is crucial to understanding the sources of inconsistent results (Abdi Tabari et al. 2024a; Allaw and McDonough 2019). Furthermore, determining the overall effectiveness of task sequencing requires a rigorous meta-analytical approach to calculate a pooled effect size (Weissgerber et al. 2021). Ultimately, assessing how different methodological choices influence the observed outcomes will provide a nuanced understanding that can guide future research and improve study designs (Plonsky and Kim 2016). Inspired by these gaps, this multi-level meta-analysis aims to critically examine empirical research on the impact of task sequencing in L2 learning. More importantly, it seeks to identify the methodological differences across existing studies and their implications, determine the overall effectiveness of task sequencing in enhancing linguistic production, and assess the extent to which these methodological variations influence the observed outcomes.

2 Background literature

2.1 Methodological variations in TS studies

The efficiency of cognitive task complexity as a sound criterion for sequencing tasks has been the main concern of numerous studies in the TBLT domain. A number of studies have been triggered by Skehan's (1998) predictions on how cognitive task complexity impacts language

performance. Skehan (1998) contends that due to the meaning-centeredness of tasks, by boosting their complexity, attention to meaning and, consequently, fluency is enhanced. Therefore, little attentional capacity is left for formal aspects, leading to a trade-off between accuracy and complexity. In other words, either accuracy or complexity is improved as a function of cognitive complexity. Other attempts have revolved around task complexity under the influence of Robinson's (2005) theoretical notion of the CH, which is believed to be the opposing view to Skehan's predictions, and the Triadic Componential Framework (TCF). The CH postulates that our attentional capacity consists of several pools of attention, rather than a single pool of attention, and there is competition within each pool rather than between pools. As a result, with a boost in cognitive task complexity, both complexity and accuracy are augmented, probably to the detriment of fluency. Given this hypothesis, Robinson (2010) offers a comprehensive framework, i.e., the TCF, for the systematic complexification of pedagogic tasks. Among the three broad categories of this framework, namely, task condition, task difficulty, and cognitive complexity, Robinson considers cognitive complexity as the main criterion for sequencing tasks. Cognitive complexity is divided into two major variables: (1) resource-directing variables and (2) resource-dispersing variables. The former involves variables, for instance, ±reasoning demands, ±here and now, and ±few elements, that attract attention to linguistic features of production while the latter comprises variables, such as ±planning time, ±single task, and ±few steps, that divert attention from these features. Most studies on task complexity have focused on the binary characteristic of the variables (e.g., +planning vs. − planning) and revolved around a dichotomy of task complexity (simple vs. complex). In their meta-analysis study on resource-directing variables considering this dichotomy, Jackson and Suethanapornkul (2013) found a small accuracy effect (Cohen's d = 0.28) and no complexity effect (d = −0.02) through time perspective whereas Malicka and Sasayama (2017) reported the highest effect sizes for lexical complexity (d = 0.34) manipulating reasoning demands and syntactic complexity (d = 0.41) using time perspective with no accuracy effects (d = 0.12 for reasoning demands and d = 0.15 for time perspective). The results from Malicka and Sasayama (2017) also indicate that accuracy (d = −0.73), fluency (d = −0.34), and structural complexity (d = −0.77) are more likely to be negatively affected by resource-dispersing variables, especially, planning, repetition, and support. In a similar vein, delving into L2 written production, Johnson (2017) reported positive small to medium effects of resource-directing variables, especially the ± here-and-now dimension, for syntactic complexity (d = 0.42), accuracy (d = 0.37), and fluency (d = 0.44) in the case of between-participant research designs and medium negative accuracy effects (d = −0.53) and medium positive lexical complexity effect (d = 0.45) in the case of within-participant research designs. Investigating resource-dispersing variables, Johnson (2017) concluded that lower cognitive pressure induced by the provision of planning time had more reliable and significant

impacts, particularly on lexical complexity (d = 0.65). He observed no conclusive evidence in favor of or against the CH as in some cases, both accuracy and complexity (e.g., between-participant research designs in resource -directing variables) were raised and in some others, there was a trade-off between them (within-participant research designs in resource-directing variables). Generally, not much evidence is available in favor of a joint rise in both accuracy and complexity. Although research into this dichotomy has substantially contributed to this line of inquiry, it is required to go beyond a conceptual simple-complex dichotomy and investigate a continuum of task versions manipulated along cognitive factors to determine the synergistic impact of cognitive complexity and gain a better account of the utility of task complexity in real classrooms (Malicka 2014a, 2018). This state of affairs has led to the reconceptualization of task complexity and the introduction of the SSRAC model.

The CH and the SSARC model have solid foundations as being grounded upon cognitive psychology and the functioning of attentional resources; nevertheless, they have been subjected to criticism (Ellis et al. 2020; Skehan 2016). The first one is that they follow a deductive approach to task sequencing and are heavily dependent on theory-driven predictions. It means that this model provides some predictions based on well-established theories and calls for experimental investigations to detect empirical evidence for these predetermined predictions. However, Skehan (2016) advocates an inductive approach to task sequencing and argues that task characteristics and conditions should initially be identified based on previous research, real-life classroom settings, and local contexts, and then the relationship between each task feature and performance should be examined so as to make generalizations and predictions about L2 learning and output (i.e., the contribution of findings in local contexts to theory). As the second drawback, in the TCF and SSARC model, resource-directing and resource-dispersing variables are vaguely distinguished. For instance, in the TCF, planning is introduced as a resource-dispersing variable, drawing learners' attention away from formal features of language; however, qualitative research (e.g., Pang and Skehan 2014) indicates that planning also has the potential to direct attention to formal aspects of language and function as a resource-directing variable. In a similar vein, there is no clear-cut distinction between different sub-variables within each of the resource-directing and resource-dispersing categories. As an example, the distinction between '±reasoning demands' and '±few elements', both belonging to the resource-directing category, is blurry. The third downside is that the deductively oriented studies based on the CH and SSARC model have offered less consistent findings as compared to the inductively oriented studies (Ellis et al. 2020). Skehan (2016) claims that studies on task characteristics, as more deductively oriented variables, provide less consistent findings than studies on task conditions, as more inductively oriented variables. Moreover, the consistency observed in some task characteristics (like task

familiarity) is due to their partial inductive basis. In a similar vein, the meta-analysis studies (e.g., Jackson and Suethanapornkul 2013; Johnson 2017; Malicka and Sasayama 2017) reveal that the resource-directing variables (associated more with task characteristics) offer less robust results than the resource-dispersing variables (associated more with task conditions). Last but not least, current research has failed to provide solid evidence in favor of the major premise of the CH and SSARC model, i.e., the joint rise of both accuracy and complexity and mainly reported a boost in either accuracy (Jackson and Suethanapornkul 2013) or complexity (Malicka and Sasayama 2017).

As Ellis et al. (2020) assert, it is too soon to evaluate the extended version of the CH, namely, the SSARC model. They claim that the formula is a bit vague and requires clarification. For instance, ‘e’ and ‘n’ standing for mental effort and a number of practice opportunities, respectively, are not elaborated sufficiently. This model also offers how to combine different variables, however, “it does not clarify how particular combinations like this would fit into a longer sequence as one would expect from a practical syllabus” (Ellis et al. 2020, p. 86). Furthermore, the proposed stabilize-simplify-automatize-restructure-complexify sequence is not adequately clarified and validated. For example, the questions as to why automatization should occur before complexification and what the next stage is after restructuring and complexification (e.g., consolidation or practice) remain unanswered.

Despite all these raised issues about the CH, it has generated plenty of research, and three meta-analysis studies (Jackson and Suethanapornkul 2013; Johnson 2017; Malicka and Sasayama 2017) have indicated that the predictions of this hypothesis are less likely to hold true. Several researchers, albeit few, have also empirically tested the claims of the SSARC model. However, no meta-analysis study has been conducted to unravel the adequacy of this model. Moreover, the studies following the SSARC model have used diverse methodological features (e.g., learner background, learning contextual factors, or interventions). Nonetheless, the impacts of task sequencing are likely to alter as a function of this methodological variation, as is the case with other areas of second language acquisition (Abdi Tabari and Cho 2022; Abdi Tabari and Miller 2021; Baralt 2014; Kim 2023; Korvesi and Michel 2022). Hence, there is a need to detect how different methodological features across TS studies bring about variations in the results.

In what follows, an overview of empirical studies with different methodological factors, including contextual factors and learner background (proficiency level, learning context [foreign language context vs. second language context], instructional contexts [face-to-face context vs. computer-mediated context]) and interventional factors (task type, modality, and task complexity operationalization), that are likely to moderate the impacts of TS on L2 production dimensions (CALF) is provided. These methodological features were selected after

a thorough content analysis of the method sections of the extracted studies. Based on this analysis, the factors shared by these studies were included. These variables were also found to be explored by several meta-analysis studies in the area of TBLT (e.g., Abdi Tabari et al. 2024b; Bryfonski and McKay 2017; Jackson and Suethanapornkul 2013; Johnson 2017; Johnson and Abdi Tabari 2022, 2023) indicating their potential to play moderating roles in the area of task sequencing.

2.2 Review of contextual factors and learner background in TS studies

2.2.1 Proficiency levels in TS studies

Learners at different levels of proficiency (i.e., elementary, intermediate, and advanced levels) go through different types of L2 processing and possess varying degrees of declarative and proceduralized knowledge, which undeniably impacts their attention allocation and L2 production (Awwad and Tavakoli 2019). Provided that high-proficient learners have richer L2 exemplar-based and rule-based systems and are more capable of carrying out parallel processing due to their greater proceduralized and automatic knowledge, they are expected to tackle the cognitive pressure imposed by task inherent features more efficiently and benefit more from TS as compared to learners with lower levels of proficiency (Awwad and Tavakoli 2019). Given the differences in high- and low-proficient learners' cognitive processes, one may contend that the claims of the CH are proficiency-dependent. In other words, high-proficient learners are probably more capable of increasing accuracy and complexity simultaneously while performing more complex tasks than their counterparts. Very few TS investigations (n = 3) have targeted elementary learners. Kim (2023) demonstrated improvements in fluency (d = 0.61) and syntactic complexity (d = 0.58) as a result of SC sequencing with no effects on accuracy (d = 0.19), suggesting that probably due to their restricted L2 lexicon and controlled processing (Kormos 2023), these learners were propelled to prioritize some aspects over others. Focusing on intermediate learners, Thompson (2014a) employed task sequencing as a treatment and exhibited evidence of increases in fluency (d = 2.08), morphological accuracy (d = 4.98), and syntactic complexity (d = 4.06) from the pretest to immediate posttest, and Lee (2021) observed promotion in lexical complexity (d = 0.87) owing to SC sequencing. Abdi Tabari and Cho (2022), as the only study on advanced learners, unraveled the greater efficiency of SC sequencing over CS and No sequencing in CALF. Malicka (2014a) investigated improvements in CALF as a function of proficiency level using SC and RAN sequencing and found that although low-proficient learners primarily attended to syntactic complexity, especially words per clause (d = 0.37) in the medium task, and high-proficient ones to accuracy (d = 0.60) in the most complex task regardless of the position of tasks in the sequences, task sequencing exerted marginal effects on CALF measures,

particularly at the low proficiency level ($0.03 < d < 0.15$). On the whole, these findings suggest that all proficiency levels may benefit from SC sequencing but in different ways as they are at different stages of L2 development. Hence, there is a need for further empirical investigations on how learners' proficiency level mediates the TS effects.

2.2.2 Learning contexts in TS studies

Contexts of learning with distinctive characteristics exert a lot of influence on language learning (Yoo and Kim 2023). Among these learning contexts are English second language (ESL) contexts and English foreign language (EFL) contexts. The differences between these two contexts are well-acknowledged in the area of language learning. In the former, learners are more exposed to linguistic and sociolinguistic aspects of the target language, get more involved in real-life tasks, and have abundant opportunities for processing, storing, and retrieving L2 daily beyond their formal classroom settings (Barrot and Gabinete 2019), which, drawing upon the usage-based theory (Tomasello 2003), expedites the automatization and development of the target language. However, in the latter, learners have limited access to the input of the target language, mainly provided within the borders of classrooms in the form of written language. One speculation is that ESL learners have to implement complex real-life tasks outside the boundaries of the classroom, which may constantly direct their attention toward specific aspects of language, especially those facilitating meaning conveyance and successful communication like fluency and complexity. EFL learners, on the other hand, are much less likely to encounter complex real-life tasks, and due to the above-mentioned characteristics of EFL contexts, formal aspects of language like accuracy are more likely to be underscored. Based on this rationale, ESL and EFL learners tend to develop different tendencies toward various aspects of language (CALF), and certain interventions are thus required to enhance a balanced development of their interlanguage. Regarding ESL learners, moving from complex to simple tasks (CS sequencing) may be of considerable help as they are more familiar with complex versions of tasks, and gradually decreasing task complexity may help them devote more attentional capacity to formal features of the language like accuracy. As for EFL learners, by a gradual increase in cognitive complexity (SC sequencing), they can build upon the simpler versions and attend to meaning conveyance and language complexification. In this way, they are progressively prepared for authentic real-life tasks. These are just theoretical speculations based on the features of the ESL and EFL learning contexts, which require empirical investigations. However, no study has exclusively focused on how EFL and ESL learners may differ when it comes to task sequencing and which task sequencing may benefit them most.

The majority of TS studies (13 out of 16 studies) have been undertaken in FL contexts. These studies have generally reported the utility of SC sequencing in promoting all or some measures of CALF (Allaw 2017; Farahanynia 2019; Kim 2023; Korvesi and Michel 2022; Lee 2021; Levkina and Gilabert 2014; Malicka 2014a; Thompson 2014a). Furthermore, SC sequencing functioned better than other sequencing types (CS or randomized sequencing) in at least one of the CALF constructs (Allaw 2017; Allaw and McDonough 2019; Farahanynia 2019; Faruji and Ghaemi 2017; Lee 2021; Levkina 2013; Levkina and Gilabert 2014; Malicka 2014a, 2018). In the SL context, positive changes in accuracy (d = 0.94) and fluency (d = 0.43) have been found as a result of SC sequencing (Korvesi and Michel 2022) but no empirical study has particularly focused on lexical and syntactic complexity. SC sequencing has led to greater improvements in CALF measures than CS sequencing (Abdi Tabari and Cho 2022; Abdi Tabari and Miller 2021). Although the findings in the SL contexts reflect positive influences of SC sequencing, due to the low number of studies, no conclusive claims can be made. These findings corroborate that SC sequencing in both SL and FL contexts has the potential to lead to gains. However, to date, no study has been devised to examine the effectiveness of SC sequencing as a function of learning contexts to reveal which learning context provides better conditions for learners' attention allocation and improvement and which constructs are more easily affected by each learning context.

2.2.3 The instructional context in TS studies

Another contextual feature is the mediation of instruction, i.e., whether the instruction is provided in the face-to-face classroom (FtF) or computer-mediated communication (CMC) context. Although since the COVID pandemic, we have witnessed the augmentation of online classes and there have been calls for investigation into online learning contexts and their impacts on task performance and task sequencing efficacy (Lai and Li 2011), there is a dearth of research on these environments in the TS area. Korvesi and Michel (2022), as the only reviewed study conducted in both environments, found that online chat led to more accurate use of present continuous (d = 0.78) and 'can' + infinitive (d = 0.23) than FtF exchanges while FtF conversations brought about higher scores in functional language use (d = 0.50) in the posttest. They also indicated positive effects of task sequencing in both contexts, suggesting the use of TS in online contexts so as to facilitate developments in FtF classrooms. There is little research on TS in online environments and inconsistent findings have been generated. Hence, to obtain a thorough picture of TS efficacy, it is crucial to explore how TS effects differ as a function of FtF and online learning contexts (Baralt 2014).

2.3 Review of interventional factors in TS studies

2.3.1 Task types in TS studies

Different task types with different communicative goals and functional requirements are highly likely to result in task-induced variations in L2 task performance (Alexopoulou et al. 2017; Ellis 2008). In the TS domain, three task types have been investigated: picture narrations (5 studies), decision-making tasks (7 studies), and picture descriptions (4 studies). Probably, these task types yield themselves more easily to TS operationalization. All the studies including a sequence of simple-medium-complex narrations or picture descriptions have demonstrated that some of the measures of each CALF construct under investigation have undergone positive changes. SC sequencing of narrations has mainly instigated learners to generate more syntactically complex (Kim 2023; Thompson 2014a, 2014b), more grammatically accurate (Korvesi and Michel 2022; Thompson 2014a, 2014b), and fluent (Kim 2023; Korvesi and Michel 2022; Thompson 2014a) output. However, lexical complexity is largely overlooked in these studies. Sequencing picture descriptions from simple to complex versions has fostered the production of more syntactically and lexically complex language with more target-like structures at a higher pace (Allaw 2017; Allaw and McDonough 2019; Levkina and Gilabert 2014). This sequencing also promoted lexical complexity, fluency (Allaw 2017; Allaw and McDonough 2019), and accuracy (Allaw 2017; Levkina 2013; Levkina and Gilabert 2014) compared to other TS types. However, no remarkable differences between different sequences were observed for syntactic complexity (Allaw 2017). The picture is a bit vague concerning decision-making tasks since in some studies, syntactic complexity (Farahanynia 2019), accuracy (Farahanynia 2019; Malicka 2014a), and lexical complexity (Farahanynia 2019; Lee 2021) have been augmented whereas, in other studies, no significant variations have been detected in syntactic complexity (Lee 2021; Malicka 2014a), fluency (Farahanynia 2019; Malicka 2014a), and lexical complexity (Malicka 2014a). SC sequencing in this task type has functioned better than other TS types (like CS or randomized sequencing) in CALF (Abdi Tabari and Cho 2022; Abdi Tabari and Miller 2021; Farahanynia 2019; Lee 2021; Malicka 2018). However, Malicka (2014a, 2014b) reported no difference between different types of TS. These findings indicate the substantial role of sequencing picture narratives and descriptives from simple to complex ones in improving L2 performance. However, further research on decision-making tasks is required to detect more robust patterns. It is also required to identify which task type in a sequence works better in promoting each CALF construct. As a gap, no study has examined how the efficiency of TS is determined by task types.

2.3.2 Modality in TS studies

Another factor that might moderate TS' efficacy is the modality of production, i.e., oral and written modes of output, because different modalities involve stimulating different underlying mechanisms and information processing (Kormos 2023). Due to the spontaneity of oral production, we generally have to go through all the stages of spoken language (i.e., conceptualization, formulation, articulation, and self-monitoring based on Levelt's [1989] model of oral production) concurrently, perform automatic processing to handle this real-time pressure and have substantial accessibility to our L2 lexicon (Abdi Tabari and Cho 2022). Conversely, in written production, we have ample time to plan (offline planning) and have more control over various stages of writing (i.e., planning, execution, and monitoring based on Kellogg's [1996] model of writing). We can even devote our undivided attention to one stage at a time. Such differences in the nature of these modalities may result in variations in learners' attention allocation and L2 production (Kim and Hwang 2022). It is expected that the complex written task in SC sequencing generates the most accurate and complex written production due to having sufficient time for planning, formulation, and monitoring (Granfeldt 2007) while the complex oral task in this sequencing makes learners produce language at a higher speed (owing to its spontaneity) with more complexity (due to the cognitive pressure). Among oral studies (n = 8), Kim (2023) found that learners' spoken language was more fluent and syntactically complex using SC sequencing while accuracy remained intact. Nonetheless, learners produced more target-like structures in some studies (Levkina and Gilabert 2014; Thompson 2014a). Among written studies (n = 7), Farahanynia (2019) observed greater levels of syntactic complexity (d = 5.98) and lexical complexity (d = 1.94) and accuracy (d = 2.05) in learners' written production as a result of SC sequencing while fluency was unaffected (d = −0.13). However, by analyzing the written output, Allaw (2017) witnessed improvements in fluency (d = 2.63). As the only study focusing on both modes, Korvesi and Michel (2022) demonstrated positive influences of SC sequencing on global accuracy (d = 1.1), task-specific accuracy (d = 1.18) and fluency (d = 0.83) in text chat and oral production (d = 0.55, d = 1.31, d = 0.28, respectively). Inconsistent findings lead to inconclusive conclusions about how each modality contributes most to each construct. This underscores the necessity of performing research on how modality moderates the effectiveness of TS.

2.3.3 Task complexity operationalization in TS studies

One of the most crucial methodological considerations in devising TS interventions that seek to develop L2 performance is how cognitive complexity is operationalized and what cognitive variables (resource-directing or resource-dispersing variables) in the TCF are considered. In the reviewed studies, the frequent resource-directing variables are ± few elements and ±

reasoning demands, and the frequent resource-dispersing variables are ±planning, ±perspective-taking, and ±task structure. Half of the studies (n = 8) have followed the principles of the SSARC model including both resource-directing and resource-dispersing variables. It can be anticipated that these studies are highly likely to provide more robust evidence for supporting or refuting the model. Among them, Abdi Tabari and Cho (2022); Abdi Tabari and Miller (2021); Allaw (2017); Allaw and McDonough (2019) and Thompson (2014a, 2014b) concluded that the results supported the SSARC model as the included measures have substantially been elevated using SC sequencing. However, Kim (2023) and Farahanynia (2019) observed no notable changes in accuracy (d = 0.13) and fluency (d = −0.13), respectively, which puts the claims of this model under question. The remaining studies (n = 8) have confined themselves to resource-directing variables. Some of them demonstrated improvements in the measures owing to SC sequencing, generally advocating the model (e.g., Faruji and Ghaemi 2017; Korvesi and Michel 2022; Levkina 2013; Levkina and Gilabert 2014; Malicka 2018). However, Malicka (2014b) claimed that task complexity, rather than task sequencing, led to better performance since there was no difference between SC and RAN sequencing and the most complex task brought about greater complexity and accuracy. In a similar vein, Lee (2021) contends that too cognitive task complexity decreased syntactic complexity as in his study, as compared to the simplest task, the medium task (d = 0.67), rather than the complex task (d = 0.22), improved this language dimension most. The divergent findings and conclusions reveal that to evaluate the predictions of this model, there is a need to operationalize task sequencing precisely based on the principles of the SSARC. It is also essential to consider other resource-directing and resource-dispersing variables in the TCF to gain a deeper insight into TS. Furthermore, investigating how different ways of operationalizing TS may moderate the results can substantially contribute to this line of research. Table 1 presents the reviewed studies' main features, their study designs, methodological features, and major findings.

Table 1: Keywords used in database searches.

Task sequencing terms	L2 production terms	TBLT terms
Task sequencing	Syntactic complexity	TBLT
Task design	Accuracy	The cognition hypothesis
Task classification	Lexical complexity	Trade-off hypothesis
The SSARC model	Fluency	Limited attentional capacity model
	L2 speaking	The triadic componential framework
	L2 writing	TBLT

3 Present study

In this study, we aim to investigate the impact of task sequencing (TS) on L2 learning using a meta-analysis approach, which is designed to synthesize previous quantitative findings to provide a comprehensive perspective. As empirical studies on task sequencing greatly differ from each other, particularly methodologically, we used a multi-level approach to account for variations not only at the sample level but also at the measurement level. In this way, we could control for methodological differences across empirical studies in estimating overall average effect sizes. Further, this allowed us to precisely estimate the extent to which these differences influenced the overall effectiveness of TS on L2 learning. Therefore, the present study was guided by the following three research questions.

RQ#1: What is the overall average effect size of TS in improving L2 learning?

RQ#1: What are the methodological differences across empirical studies on the impact of TS?

RQ#3: To what extent do the methodological differences influence the overall impact of TS on L2 learning?

4 Methods

In this section, we describe how we planned to build our dataset for the meta-analysis and to analyze the collected data to answer the research questions.

4.1 Literature search

To identify previously published empirical studies on the impact of task sequencing (TS) in L2 learning, we employed a comprehensive exploration strategy, examining numerous sources. In terms of a criterion for publication year, as our focus is mainly on the SSARC model, the date it was introduced is the starting date, i.e., 2010 (Robinson 2010). In other words, we did not consider the date and extracted all the related articles with the keywords.

First, we conducted thorough keyword searches across various databases. The first and third authors independently searched the following databases using the search terms listed in Table 1: Linguistics and Language Behavior Abstracts (LLBA), Applied Social Sciences Index and Abstracts (ASSIA), Educational Resources Information Center (ERIC), ProQuest Linguistics, ProQuest Education Journals, ProQuest Psychology Journals, ProQuest Dissertations and Theses Global, ResearchGate, Google Scholar, Google, and Listserv. We went through the abstract of the found articles in the databases quickly at the time of searching and if they were irrelevant to task sequencing, they were not included. Therefore, a total of 85 studies were returned after database searches.

Second, a total of 13 additional studies were found after journal searches. Initially, we reviewed the tables of contents of the top 50 journals in applied linguistics, as identified by the Scimago Journal and Country Rank (accessible at https://www.scimagojr.com/journalrank.php?category=1203). This process included performing keyword searches on each journal's website and scrutinizing their tables of contents. Additionally, the primary author performed targeted keyword searches within the WorldCat database using the following combinations:

> (i) "TBLT" and "L2 speaking," (ii) "TBLT" and "L2 writing," (iii) "Task sequencing" and "L2 speaking," (iv) "Task sequencing" and "L2 writing," (v) "The SSARC model" and "L2 speaking," (vi) "The SSARC model" and "L2 writing," (vii) "Limited Attentional Capacity Model" and "L2 speaking," (viii) "Limited Attentional Capacity Model" and "L2 writing," (ix) "Cognition Hypothesis" and "L2 speaking," (x) "Cognition Hypothesis" and "L2 writing," (xi) "CALF" and "L2 speaking," (xii) "CALF" and "L2 writing."

Third, we checked the references of all the extracted studies, and only one study was added. No meta-analyses have been done on task sequencing, to the best of our knowledge. There were just two meta-analyses on task complexity, and the references of both were checked.

Fourth, we conducted forward searches by extending our search to include listserv archives, ResearchGate, and Google Scholar. Recognizing the valuable insights provided by dissertations and theses due to their rigorous research designs, these sources were also included (Vuogan and Li 2023). However, no additional studies were added based on these forward searches.

Lastly, the first author utilized social media platforms to access unpublished or restricted-access theses, dissertations, and articles, ensuring a comprehensive selection of studies for our research synthesis and meta-analysis. The initial WorldCat search identified 20 notable TBLT scholars. The first author then conducted detailed investigations into each scholar's faculty webpage, while the third author examined the personal and professional web pages of these scholars. However, we could not find any additional studies. Collectively, these searches yielded 90 potential studies. Figure 1 illustrates the PRISMA flow diagram showing the entire literature search process.

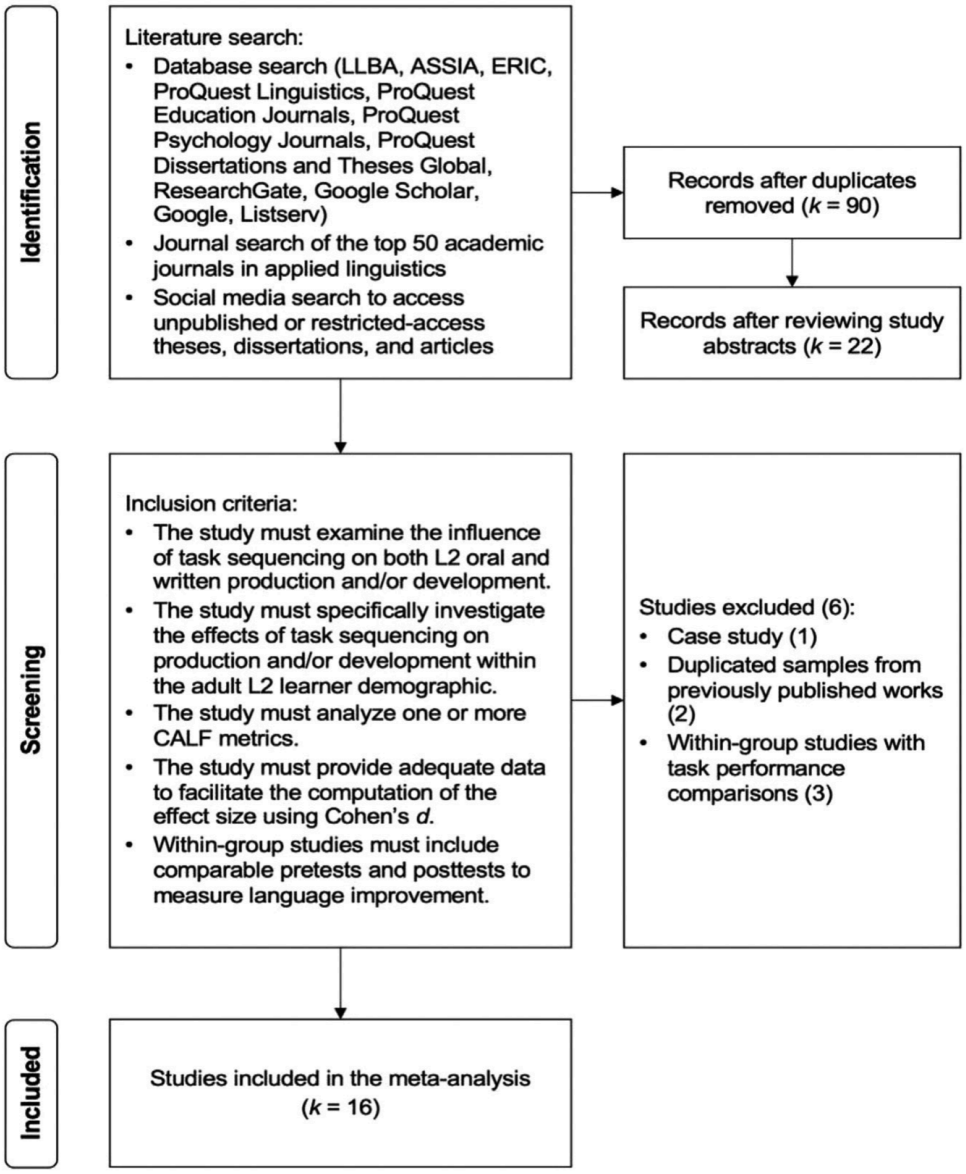


Figure 1: PRISMA flow diagram (Page et al. 2021).

To refine our selection process, the first and third authors conducted a preliminary screening of studies, applying a comprehensive set of inclusion and exclusion criteria. Publications were initially excluded if they were in languages other than English. The inclusion criteria were as follows:

1. The study must examine the influence of task sequencing on both L2 oral and written production and/or development.
2. The study must specifically investigate the effects of task sequencing on production and/or development within the adult L2 learner demographic.
3. The study must analyze one or more CALF metrics.
4. The study must provide sufficient statistical data to facilitate the computation of effect size using Cohen's *d*.

5. Within-group studies must include comparable pretests and posttests to measure language improvement.

As part of the exclusion criteria, studies were eliminated if they:

1. Did not focus on L2 oral or written production.
2. Centered on first language (L1) participant populations.
3. Lacked sufficient statistical data.

Following this initial screening of study abstracts, the pool of potential studies was reduced to 22. A subsequent review led to further exclusions:

1. Six studies were excluded, including one case study.
2. Two studies were excluded as they were duplicated samples from previously published works.
3. Three within-group studies were excluded because they featured task performance comparisons rather than comparable pretests and posttests.

This rigorous selection process resulted in a final dataset of 16 studies, which were included in our meta-analysis (*marked with an asterisk in the references*). The first and third authors independently coded all studies, and the interrater reliability was high ($\alpha = 0.96$), and any inconsistency was solved through discussions. Table 2 provides a detailed overview of these studies, including their main features, study designs, methodological details, and key findings.

4.2 Dataset construction

A total of 16 empirical studies were then analyzed to begin our dataset construction. In doing so, we found that seven studies had more than one independent sample. For example, concerning within-group design studies, Kim (2023) assigned their participants to three different (independent) experimental groups. Similarly, Abdi Tabari and Cho (2022) randomly assigned their participants to three different (independent) conditions. Regarding between-group studies, for instance, Levkina and Gilabert (2014) had two experimental groups and one control group, resulting in three comparisons (i.e., independent samples). Consequently, we could identify a total of 23 independent samples from the collected 16 studies.

Table 2: Reviewed studies along with their study designs, methodological features, and major findings.

Study identification	Method												Findings
	Study design			Sample demographic information and setting		Data collection method			Production targets				
Author Publication year Publisher	Design type	Comparisons	Methodology	Participants Age L1	Proficiency level L2 Learning context Educational setting	Modality Class/Lab FtF/CMC Task Type	Task complexity variables Task sequencing operationalizations	Groups	Complexity metrics	Accuracy metrics	Lexical complexity metrics	Fluency metrics	
Kim, N. (2023) Language Teaching Research	Between-subject within-subject pretest-treatment-posttest design	Pretest-posttest comparison	Quasi-experimental	76 (31 M; 45 F) 18 to 25 years Korean	Elementary English FL University	Oral Class FtF PN	DirV (±FElm, ±Rsn) DispV (±Pln) S (+FElm, −Rsn, +Pln with guidance) M (+FElm, −Rsn, −Pln) C (−FElm, +Rsn, +Pln with guidance) +C (−FElm, +Rsn, −Pln)	SC SCV SCC	MLT MLU C/AS DC/C MLC FreqT	E/AS EFC/C CVF	None	Syll/M DysFlu	SC sequence (without learner-dependent variables, i.e., guidance) improved syntactic complexity and speed fluency of novice learners. However, it had less influence on accuracy and dysfluency compared to SC plus guided planning. A trade-off was observed between syntactic complexity and accuracy.

Malicka, A. (2018) Language Teaching Research	Between-subject within-subject mixed design	Task performance comparison	Quasi-experimental	87 (24 M; 63 F) 19 to 31 years Spanish and Catalan	Intermediate English FL College	Oral Class FtF DN	DirVs (±Felm, ±Rsn) S (+FElm, −Rsn) M (−FElm, −Rsn) C (−FElm, +Rsn)	SC No	W/AS C/AS W/C	E/AS E/100W TLUP	GI TTR	USR PSR DysFlu	SC sequence generally elevated speech rate, accuracy, and phrasal complexity. Individual task performance generally reduced dysfluency makers and promoted subordination. Lexical complexity remained intact.
Korvesi, E. & Michel, M. (2022) Languages	Between-subject within-subject pretest-treatment-posttest design	Pretest-posttest comparison	Mixed method	15 (4 M; 11 F) The average age of 19 Miscellaneous (Syria, Iran, Gambia, Sudan, and Eritrea)	Low–intermediate English SL Language institute	Oral & Written Class CMC PN	DirV (±FElm) S (fewest elements) M (several elements) C (most elements)	SC	None	1st-grade errors 2nd-grade errors 3rd-grade errors E/50W Correct use of present continuous Correct use of existential ‘there is/are’ Correct use of can + infinitive	None	*Oral* SR ALP *Written* TC TW	SC sequence led to greater fluency and global and task-specific accuracy in both oral and written modes. Written CMI can be used to prepare students for FtF interactions.

Malicka, A. (2014a) The University of Barcelona, Spain	Between-subject within-subject mixed design	Task performance comparison	Quasi-experimental	117 19 to 31 years Spanish and Catalan	Intermediate English FL College	Oral Class FtF DM	DirVs (±Felm, ±Rsn) S (+ FElm, −Rsn) M (−FElm, −Rsn) C (−FElm, + Rsn)	No SC RAN	W/C W/AS C/AS	E/AS E/100W TLUP	DI GI	USR PSR DsyFlu	RAN sequence led to higher fluency and lexical complexity than while SC displayed greater accuracy. Both SC and RAN sequences promoted accuracy and lexical complexity, but slightly reduced fluency from simple to complex tasks. A trade-off was observed between subordination and phrasal complexity in these groups. SC sequence brought about higher accuracy, speech rate, and phrasal complexity while individual task performance led to lower disfluencies and phrasal complexity but higher subordination. No difference between groups was found in

													lexical complexity
Levkina, M. (2013) The University of Barcelona, Spain	Between-subject within-subject pretest-treatment-posttest-delayed posttest design	Pretest-posttest comparison	Experimental	61 (13 M; 48 F) Average age of 24.76 Spanish	Intermediate English FL University	Oral Class & Lab FtF PD	DirVs (±spatial Rsn, ±PersT) S (−patial Rsn, −PersT) M (+ spatial Rsn, −PersT) C (+ spatial Rsn, + PersT)	SC RAN	None	NCE NT NCT	None	None	SC and RAN sequences improved all measures merely in a laboratory context, but SC led to a greater improvement from pretest to delayed posttest, showing the greater effectiveness of this treatment in the long-term retention of information, which corroborates the SSARC model.
Levkina, M. & Gilabert, R. (2014) Bloomsbury	Between-subject within-subject pretest-treatment-posttest-delayed posttest design	Pretest-posttest comparison	Experimental	43 18 to 23 years Spanish	Intermediate English FL University	Oral Class & Lab FtF PD	DirVs (±spatial Rsn, ±PersT) S (−patial Rsn, −PersT) M (+ spatial Rsn, −PersT) C (+ spatial Rsn, + PersT)	SC CS RAN	None	STE CSTE	None	None	SC, CS, and RAN sequences significantly improved knowledge of target spatial expressions in the short and long run; the CS sequence exhibited greater improvements in the long run than the RAN sequence.

Malicka, A. (2014b) Bloomsbury	Between-subject within-subject mixed design	Task performance comparison	Experimental	50 19 to 25 years Spanish and Catalan	Intermediate English FL University	Oral Class FtF DM	DirVs (±Felm, ±Rsn) S (+FElm, −Rsn) M (−FElm, −Rsn) C (−FElm, +Rsn)	SC RAN	W/C W/AS	E/AS E/100W	DI GI	USR PSR DsyFlu	Task complexity led to a simultaneous increase in complexity and accuracy to the detriment of fluency, confirming the prediction of the Cognition hypothesis. However, the SC sequence did not yield greater qualitative changes than the RAN sequence, indicating the overriding impact of cognitive complexity, as compared to task sequencing, on task performance.
Thompson, C. (2014b) Bloomsbury	Between-subject within-subject pretest-treatment-posttest-delayed posttest design	Pretest-posttest comparison	Quasi-experimental	24 (8 M; 16 F) 19 to 21 years Japanese	Intermediate English FL University	Oral Class FtF PN	DirV (±intentional Rsn) DispV (±Pln) S (least intentional Rsn, most Pln time) M (more intentional Rsn, less Pln time) C (most intentional Rsn, least Pln time)	SCG SCU	C/AS	None	None	None	SC sequence with guided or unguided planning improved syntactic complexity (subordination).

Farahanynia, M. (2019) Tabriz Islamic University, Iran.	Between-subject within-subject mixed design	Task performance comparison	Experimental	80 (33 M; 47 F) 23 to 35 years Persian	Intermediate English FL Language institute	Written Class FtF DM	DirV (±Rsn) DispV (±Pln) S (+ Pln, −Rsn) M (-Pln & −Rsn) C (−Pln & + Rsn)	SC RAN	C/AS	EFC	GI	SR	SC and RAN sequences raised accuracy, syntactic complexity, and lexical diversity but left fluency unaffected. The RAN sequence led to greater improvements in all measures but accuracy than the SC sequence. There was a trade-off between accuracy and fluency.
Faruji, L. F., & Ghaemi, F. (2017) Journal of Teaching Language Skills	Between-subject within-subject pretest-treatment-posttest design	Pretest-posttest comparison	Quasi-experimental	90 20 to 32 years Persian	Intermediate English FL University	Written Class FtF PN	DirV (±intentional Rsn) S (least intentional Rsn) M (more intentional Rsn) C (most intentional Rsn)	SC RAN CG	None	EFT	None	None	SC group displayed better performance in accuracy RAN and Control groups.

Allaw, E. (2017) Concordia University, Canada	Between-subject within-subject pretest-treatment-posttest-delayed posttest design	Pretest-posttest comparison	Quasi-experimental	42 (18 M; 24 F) The average age of 13.32 Arabic	Elementary French FL Elementary school	Written Class FtF PD	DirV (±spatial Rsn) DispV (±task structure) S(+ spatial Rsn, +task structure) M (+ spatial Rsn, −task structure) C (−spatial Rsn, −task structure)	SC CS RAN	C/W	C/C	NSE	W/30min	Task complexity improved all measures, confirming the cognition hypothesis. SC sequence led to more accurate and fluent output than CS sequence, especially immediately after treatment. SC and CS sequences showed similar results regarding complexity, indicating that task sequencing is irrelevant to syntactic complexity.

Lee, J. (2021) The Journal of Asia TEFL	Between-subject within-subject mixed design	Task performance comparison	Quasi-experimental	88 (59 M; 29 F) Korean	Low-intermediate English FL University	Written Class FtF DM	DirV (±FElm) S (fewest elements) M (several elements) C (most elements)	SC CS	MLT C/T Nominals/T Ajectivals/T Adverbials/T	None	TTR GI AWL K1 tokens K2 tokens Off-list tokens	None	Increasing task complexity to a limited extent brings about higher syntactic complexity, as most improvement was evident in medium tasks in both SC and CS sequences. SC sequence generally improved global measures while CS elevated local measures. CS sequence generally generated a more diverse vocabulary than the SC sequence. Moreover, the higher the task complexity level, the greater the lexical diversity in both sequences.

Abdi Tabari, M., & Cho, M. (2022) Language Teaching Research	Between-subject within-subject mixed design	Task performance comparison	Experimental	100 (44 M; 56 F) 18 to 20 years Various L1 background	Advanced-mid English SL University	Written Class FtF DM	DirV (±FElm) DispV (±Pln) S (+ FElm, + Pln) M (+ FElm, −Pln) C (−FElm, −Pln)	SC CS No	MLC CN/T	EFC/C CVF	MTLD WRDF	W/M	SC sequence improved syntactic complexity and accuracy over time more than the CS sequence. SC and CS sequences each promoted measures more than individual task performance, especially in medium and complex tasks, supporting the claims of the SSARC model.
Abdi Tabari, M., & Miller, M. (2021) Canadian Journal of Applied Linguistics	Between-subject within-subject mixed design	Task performance comparison	Experimental	90 (44M; 46 F) The average age of 20.6 Various L1 background	Upper-intermediate English SL University	Written Class FtF DM	DirV (±FElm) DispV (±Pln) S (+ FElm, + Pln) M (+ FElm, −Pln) C (−FElm, −Pln)	SC No	MLT DC/T CN/T CN/C	EFC/C CVF	MTLD WRDF	W/T	SC sequence elevated syntactic and lexical complexity, accuracy, and fluency more than individual task performance, supporting the theoretical claim of the SSARC model.

Allaw E. & McDonough K. (2019) System	Between-subject within-subject pretest-treatment-posttest-delayed posttest design	Pretest-posttest comparison	Quasi-experimental	42 (18 M; 24 F) 13 to 14 years Arabic	Elementary French FL Elementary school	Written Class FtF PD	DirV (±spatial Rsn) DispV (±task structure) S(+ spatial Rsn, +task structure) M (+ spatial Rsn, −task structure) C (−spatial Rsn, −task structure)	SC CS RAN	None	C/C	NSE	W/30min	SC and CS sequencing improved all measures over time. SC sequence led to more accurate, lexically complex, and fluent output than CS sequence, especially in the immediate posttest.
Thompson C. (2014a). Josai International University, Japan	Between-subject within-subject pretest-treatment-posttest-delayed posttest design	Pretest-posttest comparison	Quasi-experimental	24 18 to 22 years Japanese	Upper-intermediate English FL University	Oral Class FtF PN	DirV (±intentional Rsn) DispV (±Pln) S (least intentional Rsn, most Pln time) M (more intentional Rsn, less Pln time) C (most intentional Rsn, least Pln time)	SCG SCU	C/AS	EFC	None	PSR	SC sequencing with or without guided planning improved syntactic complexity, fluency, and morphological accuracy.

Furthermore, we found that the identified 23 samples could be categorized into four types: (1) pretest-posttest within-group, (2) pretest-delayed posttest within-group, (3) posttest between-group, and (4) delayed posttest between-group, based on their use of measurements at different time points and for various group comparisons. For example, Levkina (2013), Levkina and Gilabert (2014), Allaw (2017), and Allaw and McDonough (2019) employed all four types, utilizing pretest, posttest, and delayed posttest measures for both treatment and control groups. In contrast, Kim (2023) and Korvesi and Michel (2022) employed pretest and posttest measures for single groups, resulting in just one type – pretest-posttest within-group. Consequently, we had (1) 14 pretest-posttest within-group samples (N = 343), (2) nine pretest-delayed posttest within-group samples (N = 219), (3) 15 posttest between-group samples (N = 741), and (4) five delayed posttest between-group samples (N = 171), totaling 52 independent samples (N = 1,474).

4.3 Coding scheme

The purpose of this study is to code for methodological differences across empirical TS studies, as discussed in RQ1. To provide a strong rationale for our coding scheme, we connect each coded factor to the predictions of the *Cognition Hypothesis* (Robinson 2001) and *SSARC* (Robinson 2010) which posit that task complexity influences cognitive processing, which in turn affects language learning outcomes. The following sections explain how each factor in the coding scheme aligns with these predictions.

First, we coded contextual factors and learner background variables, such as participants' L2 proficiency level, learning context (e.g., EFL, ESL), and instructional context (e.g., face-to-face/computer-mediated instruction). These factors are important because, according to the *Cognition Hypothesis*, learner proficiency and the learning environment are crucial in determining the cognitive load experienced during tasks. For example, higher proficiency learners might process complex tasks more efficiently, while the learning context (e.g., face-to-face vs. computer-mediated) can affect task engagement and cognitive demands. By coding for these variables, we can understand how these background factors interact with task complexity to influence cognitive load and learning outcomes.

Second, we coded for interventional factors, such as task type (e.g., narration, description, decision-making), modality (e.g., written, oral), and task complexity (e.g., resource-directing (±few elements), resource-dispersing (±planning)). These task-related factors are key to the *Cognition Hypothesis*, which predicts that tasks with varying cognitive demands (e.g., more elements to process, more planning required) will engage different cognitive processes. For instance, tasks with higher cognitive complexity are thought to demand more attention and cognitive resources, potentially leading to more language output or better language

development, depending on how well learners manage these demands. By coding these variables, we aim to examine how methodological differences in task complexity – such as the cognitive load imposed by different types of tasks – affect learning outcomes.

Third, for sample demographics, we coded variables such as the number of participants, their age, their first language (L1), and their target language. These demographic details help explain how individual differences in language experience and cognitive resources might influence task performance and learning outcomes. For instance, older learners or learners with higher L2 proficiency may approach tasks with different cognitive strategies compared to younger or less proficient learners. Understanding these demographic factors allows us to examine how learner characteristics intersect with task complexity to shape cognitive engagement and task performance, in line with the *Cognition Hypothesis* and *SSARC*.

Lastly, we coded the measurement criteria used in the studies, specifically whether syntactic/lexical complexity, accuracy, and/or fluency metrics were used. These outcome measures are directly linked to the *Cognition Hypothesis* and *SSARC*, which suggest that complex tasks, by increasing cognitive load, may lead to greater improvements in linguistic complexity (e.g., syntax and lexicon) and accuracy. By coding for these metrics, we can assess how different task characteristics (such as complexity and modality) influence the specific aspects of language learning that are measured across studies.

Table 3 summarizes the list of these independent variables of interest for this coding phase. Each variable has been selected based on its potential to affect cognitive load and language learning outcomes, following the predictions of the *Cognition Hypothesis* and *SSARC*. This coding scheme provides a clear framework for understanding how methodological differences in task design impact cognitive processes and, ultimately, language acquisition.

4.4 Quality assessment

In reviewing the samples to code independent variables, we further conducted a quality assessment to ensure that the remaining samples for the current meta-analysis have methodological rigor sufficient to minimize potential biases. Among many different ways of quality check for a meta-analysis, we followed Vuogan and Li's (2023) suggestion, such that each sample needs to be checked "whether the independent variable was strictly controlled, whether the studies were comparable, and whether instructional treatments aligned with theories" (p. 18). To this end, the authors of the current study agreed to choose three categories, such as theoretical accounts, data collection method, and data analysis method and came up with 10 criteria for assessing the quality of the samples. If a sample scores higher than 8 out of 10, we conclude that the sample meets the requirements for our quality assessment.

Table 3: Coding strategies.

Category		Independent variable
Target moderator	Contextual factors and learner background	L2 proficiency levels
		Learning context (e.g., EFL, ESL)
		Instructional context (e.g., Face-to-face/computer-mediated instruction)
	Interventional factors	Task type (e.g., narration, description, decision-making)
		Modality (e.g., written, oral)
		Task complexity (e.g., resource-directing (±few elements), resource-dispersing (±planning))
Background	Study identification	Publication Type
		Publication source
	Sample demographic information	No. of participants
		Participants' age
		First language
		Target language
Measurement	Target of instruction	Syntactic complexity metrics
		Lexical complexity metrics
		Accuracy metrics
		Fluency metrics

The 10 criteria used for evaluating the studies were as follows: First, for theoretical accounts, we examined (1) how task sequencing was defined, (2) whether an appropriate theoretical framework was adopted, and (3) whether relevant research questions were presented. Second, for the data collection methods, we evaluated (4) whether relevant metrics were measured, (5) how task sequencing was operationalized, (6) whether the cognitive load associated with simple and complex task versions was validated to ensure alignment with theoretical predictions (e.g., confirming that complex tasks imposed greater cognitive demands as proposed by the Cognition Hypothesis), and (7) whether an appropriate study design type was adopted. Third, for the data analysis methods, we assessed (8) whether reliability checks were conducted, (9) whether the normality of data was verified, and (10) whether appropriate statistical tests were employed. As these criteria were well aligned with our inclusion criteria, we found that all samples adequately satisfied all criteria (i.e., total score ≥ 8). See Table 4 for more details about examples of how we coded each variable as well as the results of the quality assessment for the 23 samples.

Table 4: Quality assessment strategies.

Category	Independent variable	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12	S13	S14	S15	S16	S17	S18	S19	S20	S21	S22	S23
Theoretical accounts	(1) Task sequencing definition (i.e., sequencing tasks based on cognitive task complexity)	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√
	(2) Theoretical framework (e.g., Robinson's SSARC model, Robinson's triadic componential framework, Robinson's cognition hypothesis, Kormos' bilingual model of oral production, Skehan's trade-off hypothesis, Skehan's limited attentional capacity, Skehan's task difficulty, Levelt's model of production, Dynamic usage-based (DUB) approaches, Form–use–meaning mappings)	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√
	(3) Research questions (Relevant/Irrelevant)	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√
Data collection method	(4) Complexity metrics (e.g., syntactic complexity, accuracy, lexical complexity, fluency)	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√
	(5) Task sequencing operationalization (i.e., Simple -> Medium -> Complex; e.g., Task 1: Simple (+few elements, −reasoning) Task 2: Medium (−few elements, −reasoning) Task 3: Complex (-few elements, +reasoning)	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√
	(6) Cognitive load verification (e.g., Affective variables questionnaire (AVQ), Time judgment task (TJT), Stimulated recall (SR), Task difficulty questionnaire)				√	√	√	√	√	√	√			√	√	√	√		√	√	√	√	√	√	√
	(7) Methodology (e.g., Quasi-experimental, Experimental)	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√
Data analysis method	(8) Reliability checking (Yes/No)	√	√	√	√		√	√	√	√	√	√	√				√	√	√	√	√	√		
	(9) Normality checking (Yes/No)	√	√	√	√		√	√	√	√	√	√	√		√	√	√	√			√	√	√	√
	(10) Statistical tests (e.g., One-way ANOVA, MANOVA, Paired sample *t*-tests, Mann–Whitney *U*-test, Wilcoxon signed-rank tests, Kruskal-Wallis tests, Two-way repeated-measures ANOVAs, Repeated meaures ANOVA, Independent samples *t*-tests)	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√
Total scores		9	9	9	10	8	10	10	10	10	10	10	10	8	9	9	10	9	9	9	10	10	9	9

4.5 Effect size calculation

After checking the quality of the identified samples, we extracted descriptive statistics of each sample to calculate effect sizes. We used Hedges' *g* (Hedges 2019) as a unit of our effect size as this approach is more reliable for small sample sizes than Cohen's *d*, which is one of the widely used units in the field (Cohen 1988). Calculating effect sizes is to convert the treatment effects reported by each sample which used various types of measurement in different scales into a comparable estimate by standardizing the mean differences between the pretest and (delayed) posttest of a single group (i.e., within-group) or between (delayed) posttest results of two different groups (i.e., between-group). In this way, within-group effect sizes indicate a degree of *improvement* in participants' performance after TS tasks, while between-group effect sizes represent *treatment effects* of TS tasks by comparing performances between those who received treatments and those who did not. Concerning how to interpret the size of effect sizes, it is widely accepted that 0.2, 0.5, and 0.8 are for small, medium, and large effect sizes for the standardized mean differences between two groups in the field of social science. In the present study, we adopt Plonsky and Oswald's (2014) effect size interpretation benchmarks, which were suggested particularly for L2 studies. According to their suggestions, 0.6, 1.0, and 1.4 are small, medium, and large effect sizes for within-group comparisons, and 0.4, 0.6, and 0.9 are for between-group comparisons. As a result, we calculated (1) 71 pretest-posttest within-group effect sizes, (2) 18 pretest-delayed posttest within-group samples, (3) 120 posttest between-group effect sizes, and (4) 10 delayed posttest between-group effect sizes, totaling 219 effect sizes.

4.6 Outlier detection

Similar to the quality assessment phase, after calculating effect sizes, we employed the *metafor* package (Viechtbauer 2010) in R software to detect potential outliers and influential cases. Unlike other methods for this purpose, the *metafor* package offers a mathematical approach to identify outliers and influential cases based on eight different standards (refer to the *metafor* package webpage for details on these standards). Identifying these cases is crucial as they can disproportionately impact subsequent data analyses, such as estimating overall average effect sizes and conducting meta-regressions. Therefore, outlier detection is essential for conducting a reliable meta-analysis. We conducted outlier detection four times for each of the four sample types, categorized based on the timing of measurements and comparisons across different groups. To maintain data integrity, we opted to exclude the identified outliers from subsequent data analyses (see Figure 2).

The results revealed that (1) 14 samples for within-group effect sizes showed no signs of outliers (i.e., remained *k* (number of samples) = 14, remained *n* (number of effect sizes) = 71), (2) nine samples for within-group delayed effect sizes had one outlier (i.e., remained *k* = 8, remained *n* = 17), (3) 15 samples for between-group effect sizes showed no signs of outliers (i.e., remained *k* = 15, remained *n* = 120), and (4) five samples for between-group delayed effect sizes had two outliers (i.e., remained *k* = 3, remained *n* = 4). It should be noted here that due to their small number of samples; we opted not to conduct further data analyses (i.e., computing overall average effect sizes and employing moderator analyses) for the second (i.e., samples for within-group delayed effect sizes) and fourth sample types (i.e., samples for between-group delayed effect sizes).

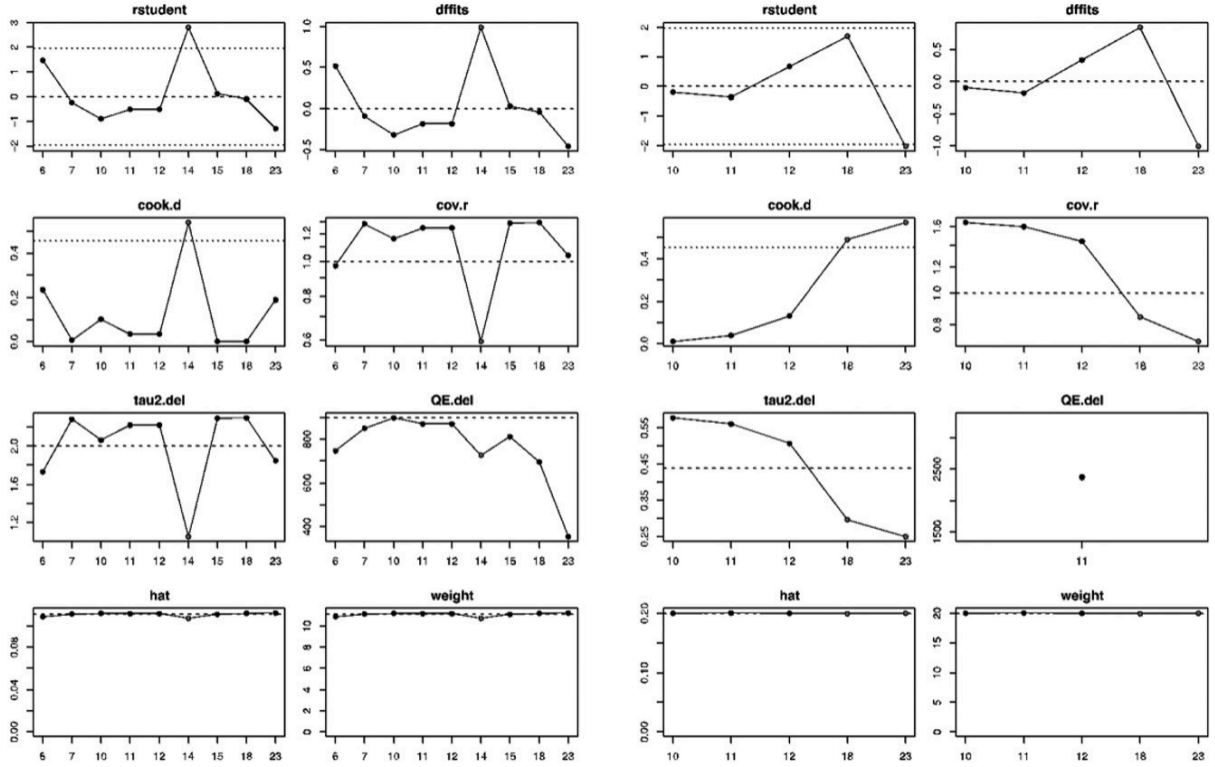


Figure 2: Results of outlier detection by metafor package.

4.7 Data analysis plan

The follow-up data analyses focused on computing overall average effect sizes and conducting meta-regressions to address the second and third research questions. We used Stata version 16.1 (StataCorp 2020) software as the statistical tool. First, to compute the overall average effect size from the calculated effect sizes nested within each independent sample, we performed a multi-level regression analysis using the *meglm* library in Stata, incorporating standard errors of the effect sizes as the known variance. This approach allows us to give more weight to effect sizes from larger samples (i.e., lower standard errors) because they offer more

precise results. Second, by including moderator variables in the regression equation, we investigated whether these variables significantly influenced the strength or direction of the overall effectiveness of the treatment strategy computed in the previous step. Additionally, we included control variables to account for other factors that might influence the outcome, providing a clearer understanding of both the overall average effect size and moderating effects. The list of moderator and control variables for each analysis is presented in the results section as part of the first research question. It should be noted here that all data analyses were conducted for each sample type, such as within-group effect sizes and between-group effect sizes.

5 Results

5.1 RQ #1: Identified methodological differences across empirical studies

While our coding strategies initially included a comprehensive list of moderator variables and values of our interests, it is important to note that not all variables and values could be coded due to limitations in data availability across studies. Despite our efforts to systematically collect data, some studies did not report certain variables or provided incomplete information, resulting in a subset of variables being identified as part of methodological differences across empirical studies. As such, the identified moderator variables and their corresponding values represent those available from the included studies, as shown in Table 5, and we finally included seven moderator variables to answer the first research question.

First, we coded learners' L2 proficiency levels into three values: low, intermediate, and advanced. However, the within-group effect size samples did not include participants with advanced L2 proficiency levels. Furthermore, in the between-group effect size samples, there were only two samples that included participants with low L2 proficiency levels ($k = 2$, $n = 6$). Second, the learning contexts comprised one where English is learned as a foreign language (i.e., EFL) and another where English is learned as a second language (i.e., ESL). Third, we found that TS tasks were delivered in either written or oral modalities. Fourth, there were three types of TS tasks, such as narration, description, and decision-making. The within-group effect size samples did not include the description type TS tasks, and the between-group effect size samples did not have the narration type TS tasks. Fifth, task complexity was categorized as less complex and more complex. For instance, the former encompassed resource-directing tasks with few elements, whereas the latter involved both resource-directing and resource-dispersing tasks, including planning components. Sixth, the delivery method of TA tasks, whether face-to-face or computer-mediated, was coded. Seventh, there were four constructs as

we hypothesized: syntactic complexity, lexical complexity, accuracy, and fluency. However, only one sample targeted learners' lexical complexity (k = 1, n = 2) within the within-group effect size samples.

Table 5: Identified methodological differences (moderator variables).

Moderator variable	Value	Within-group ES		Between-group ES	
		k	***n***	***k***	***n***
L2 proficiency	Low	5	39	2	6
	Intermediate	9	32	11	72
	Advanced	0	0	2	42
Learning context	EFL	13	53	12	69
	ESL	1	18	3	51
Modality	Written	11	54	7	46
	Oral	4	17	8	74
Task type	Narration	8	59	0	0
	Description	0	0	9	108
	Decision-making	6	12	6	12
Task complexity	Resource-directing	5	24	9	59
	Resource-directing + resource-dispersing	9	47	6	61
Instructional context	Face-to-face	12	69	8	25
	Computer-mediated	2	2	7	95
Construct	Syntactic complexity	7	23	10	34
	Lexical complexity	1	2	10	31
	Accuracy	12	33	14	34
	Fluency	7	13	9	21

Note. In addition to the moderator variables, we could code two control variables: publication type and methodology. The former had four values, such as journal articles, book chapters, dissertations, and conference presentations. The latter had two values based on whether the participants were randomly assigned, such as quasi-experiment and experiment.

5.2 RQ #2: Overall average effect size of TS in improving L2 learning

To answer the second research question, we computed the overall average effect sizes for each type of sample by employing multi-level meta-regression analyses without moderator variables. Initially, we identified a large effect size of 1.85 (SE = 0.46, $p < 0.001$) for within-group samples, according to Plonsky and Oswald's L2 effect size benchmark. Subsequently, for between-group samples, we found a marginal effect size of 0.30 (SE = 0.16, $p < 0.001$) based on the same benchmark. To refine our estimates with higher precision, we reran the analyses by including control variables. This revealed a medium effect size of 1.15 (SE = 0.17, $p < 0.001$) for within-group samples and a small effect size of 0.44 (SE = 0.10, $p < 0.001$) for between-group samples. In summary, the results from the within-group samples indicate a

significant improvement in learners' L2 performance through the provision of TS tasks. Further, the findings from the between-group samples suggest that providing TS tasks yields a significant treatment effect in enhancing learners' L2 performance (see Table 6).

5.3 RQ #3: moderating effects of the methodological differences

To address the third research question, we conducted multi-level meta-regression analyses with moderator variables. Initially, we constructed simple meta-regression models, with each moderator variable included in individual models to identify its moderating effects. Subsequently, we built multiple meta-regression models using the moderator variables that demonstrated statistically significant moderating effects in the previous phase. This strategy enabled us to obtain more precise estimates for each moderator variable while holding other important moderator variables constant. We applied this approach to both within-group effect size samples and between-group effect size samples, respectively (see Table 7).

Table 6: Overall average effect size estimates.

Category		**Overall average estimate**			
		k	***n***	**ES**	**SE**
Within-group samples (*N* = 343)	(w/o ctrl var.)	14	71	1.85***	0.46
	(w/ctrl var.)			1.15***	0.17
Between-group samples (*N* = 741)	(w/o ctrl var.)	15	120	0.30***	0.16
	(w/ctrl var.)			0.44***	0.10

***$p < 0.001$. *Note*. In the analysis of within-group samples, only the publication type variable was included as a control, whereas for the between-group samples analysis, both publication type and methodology were included as control variables.

Table 7: Coefficients from moderator analysis.

Moderator variable	**Value**	**Within-group sample (*N* = 343)**				**Between-group sample (*N* = 741)**			
		Simple		**Multiple**		**Simple**		**Multiple**	
		Coef.	**SE**	**Coef.**	**SE**	**Coef.**	**SE**	**Coef.**	**SE**
L2 proficiency	Low	Ref.	–	Ref.	–	Ref.	–	Ref.	–
	Intermediate	−2.06*	0.75	−1.22	0.92	−1.09**	0.37	0.06	0.81
	Advanced	Na.	–	Na.	–	−0.50	0.48	0.51	0.77
Learning context	EFL	Ref.	–	–	–	Ref.	–	–	–
	ESL	−0.60	1.05	–	–	1.07	0.73	–	–
Modality	Written	Ref.	–	Ref.	–	Ref.	–	Ref.	–
	Oral	0.96*	0.46	0.36	0.53	1.04**	0.32	−0.25	0.34
Task type	Narration	Ref.	–	–	–	Na.	–	–	–
	Description	Na.	–	–	–	Ref.	–	–	–
	Decision-making	0.75	0.55	–	–	0.33	0.30	–	–
Task complexity	Resource-directing	Ref.	–	Ref.	–	Ref.	–	Ref.	–
	Resource-directing + resource-dispersing	2.89**	1.02	2.27*	1.03	1.05**	0.33	1.35*	0.68-
Computer-mediated	Face-to-face	Ref.	–	–	–	Ref.	–	–	–
	Computer-mediated	0.04	1.09	–	–	−0.05	0.25	–	–

Moderator variable	Value	Within-group sample (*N* = 343)				Between-group sample (*N* = 741)			
		Simple		Multiple		Simple		Multiple	
		Coef.	SE	Coef.	SE	Coef.	SE	Coef.	SE
Construct	Syntactic complexity	Ref.	–	–	–	Ref.	–	Ref.	–
	Lexical complexity	−1.02	1.09	–	–	0.12	0.19	0.09	0.19
	Accuracy	0.12	0.45	–	–	0.65**	0.20	0.66**	0.19
	Fluency	−0.39	0.51	–	–	0.17	0.23	0.17	0.22

Note. Control variables were included for moderator analyses. $^{**}p < 0.01$, $^{*}p < 0.05$.

In the within-group effect size samples, our analysis revealed that the L2 proficiency (β = −2.06, SE = 0.75), modality (β = 0.96, SE = 0.46), and task complexity (β = 2.89, SE = 1.02) variables demonstrated statistically significant moderating effects in their respective simple meta-regression models. However, when these moderators were collectively included in a multiple meta-regression model, only the task complexity variable (β = 2.27, SE = 1.03) maintained its statistical significance. This result suggests that providing more complex TS tasks had a greater educational impact on the improvement of learners' L2 performance compared to less complex tasks while holding the other variables constant.

Regarding the between-group effect size samples, the construct variable (β = 0.65, SE = 0.20), along with the previously identified three variables – L2 proficiency (β = −1.09, SE = 0.37), modality (β = 1.04, SE = 0.32), and task complexity (β = 1.05, SE = 0.33) – were found to be statistically significant in their respective simple meta-regression models. Once again, when these four moderators were included in a multiple meta-regression model, both the task complexity (β = 1.35, SE = 0.68) and construct (β = 0.66, SE = 0.19) variables retained their statistical significance. Particularly noteworthy is the moderating effect of the construct variable, where a greater treatment effect of TS tasks was observed for enhancing learners' accuracy compared to other constructs such as syntactic complexity, lexical complexity, and fluency. This finding underscores the importance of providing more complex TS tasks and focusing on improving learners' accuracy, as it significantly influenced the treatment effects of TS tasks.

To facilitate readers' understanding of the findings presented in Table 6, which displays coefficients for moderator variables indicating statistically significant moderating effects (i.e., mean differences between average effect size estimates for values within moderators), Table 8 presents the average effect size estimates computed for each value of the significant moderators. This allows us to intuitively grasp the extent to which TS tasks impact learners' L2 performance, influenced by the values of the significant moderators.

Table 8: Average effect size estimates by significant moderators.

Sample	Moderator	Value	Average ES	SE
Within-group samples (N = 343)	Task complexity	Resource-directing	−0.36	0.70
		Resource-directing + resource-dispersing	1.91***	0.38
Between-group samples (N = 741)	Task complexity	Resource-directing	−0.24	0.36
		Resource-directing + resource-dispersing	1.11***	0.35
	Construct	Syntactic complexity	0.21	0.14
		Lexical complexity	0.30*	0.15
		Accuracy	0.87***	0.14
		Fluency	0.38*	0.18

Note. Average effect size estimates were obtained from the results of multiple meta-regressions for each sample, whose detailed results are shown in Table 6. ***$p < 0.001$, *$p < 0.05$.

When examining the within-group effect size samples, the results indicate that offering more complex TS tasks (e.g., resource-directing + resource-dispersing) yielded an average effect size of 1.91 (SE = 0.38), categorized as large, while providing less complex TS tasks (e.g., resource-directing only) showed statistically null impact (d = −0.36, SE = 0.70) on the improvement of learners' L2 performance.

In the analysis of the between-group effect size samples, this pattern persisted. Delivering more complex TS tasks (e.g., resource-directing + resource-dispersing) resulted in an average effect size of 1.11 (SE = 0.35), classified as medium. Conversely, providing less complex TS tasks (e.g., resource-directing only) showed a statistically null impact (d = −0.24, SE = 0.36) on the treatment effects of TS tasks on learners' L2 performance. Furthermore, focusing on enhancing learners' accuracy through TS tasks yielded an average effect size of 0.87 (SE = 0.14), also labeled as medium. However, the treatment effects of TS tasks were not evident for other constructs; average effect sizes were marginal for fluency (d = 0.38, SE = 0.18) and lexical complexity (d = 0.30, SE = 0.15), and null for syntactic complexity (d = 0.21, SE = 0.14).

6 Discussion

Regarding RQ1 concerning the overall average effect size of TS in improving L2 learning, the substantial effect size observed in within-group samples suggests a significant improvement in learners' L2 performance due to TS. This result can be attributed to several factors. Within-group studies often offered more controlled conditions and consistent task exposure, facilitating a clearer observation of the effects of TS on individual learners. The controlled environment minimized external variables and allowed for a focused examination of the

specific impact of TS, thereby resulting in higher effect sizes. Additionally, learners in within-group studies might benefit from repeated practice and feedback, integral components of TS, leading to more pronounced improvements in L2 task performance (Robinson 2010).

On the contrary, between-group samples exhibited smaller effect sizes. The initial marginal effect size of 0.30, which increased to a small effect size of 0.44 after including control variables, suggests a more modest yet significant improvement in L2 performance due to TS. This difference in effect sizes between within-group and between-group studies can be attributed to several reasons. Between-group studies inherently involved comparisons between different cohorts, introducing variability in learner characteristics, teaching methods, and learning environments. These differences can dilute the observed effects of TS. Furthermore, between-group studies often had less control over external factors, such as instructional variations, leading to lower effect sizes compared to the more controlled within-group studies (Mackey and Gass 2016).

The inclusion of control variables in the analyses helped refine the effect size estimates by accounting for potential confounding factors. The medium effect size of 1.15 for within-group samples and the small effect size of 0.44 for between-group samples indicate that while TS had a substantial impact on individual learners in a controlled setting, its effect was more modest when applied across different groups and settings. This underscores the importance of context and control in measuring the effectiveness of instructional interventions like TS (Ellis et al. 2020). Overall, the results from within-group samples revealed a significant improvement in learners' L2 performance through the provision of TS in a simple-to-complex order. The findings from between-group samples imply that sequencing pedagogic tasks in this manner yielded a significant treatment effect, albeit smaller, in enhancing learners' L2 performance. These differences underscore the importance of study design and context in examining the impact of instructional interventions, emphasizing the need for careful consideration of methodological factors in future research (Plonsky 2017).

The results obtained for RQ1, which focused on identifying methodological differences across empirical studies, can be explained by considering several key reasons. One notable finding was the uneven distribution of L2 proficiency levels among the samples, with no advanced proficiency participants in within-group studies and only two low proficiency samples in between-group studies. This disparity may stem from the challenges of recruiting advanced L2 learners or the preference of researchers to study intermediate-level learners who constitute the largest demographic in language programs (Vuogan and Li 2023). Additionally, advanced learners might be underrepresented in within-group studies due to their potentially less varied performance, which reduces the perceived need for detailed intra-group comparisons (Plonsky and Kim 2016). Regarding the learning contexts, the majority of studies were conducted in

EFL settings, with fewer in ESL contexts. This discrepancy is likely because EFL contexts are more prevalent globally, leading to a higher number of studies in these settings (Ortega 2009; Rose et al. 2021). ESL contexts, being more specific to certain regions such as English-speaking countries, might offer fewer opportunities for large-scale research (Barrot and Gabinete 2019).

The modality of task sequencing tasks also varied, with more written modality tasks in within-group studies and more oral modality tasks in between-group studies. Written tasks might be easier to control and analyze within groups due to their static nature. In contrast, oral tasks, being dynamic and interactive, are often used in between-group studies to explore a broader range of performance outcomes across different participant groups (Kormos 2023). The types of task sequencing tasks differed, with within-group samples lacking description tasks and between-group samples lacking narration tasks. Task type selection likely reflects the specific research aims and the context in which the study was conducted (Alexopoulou et al. 2017). For example, narration tasks might be favored in studies focusing on continuous speech, while description tasks are used in contexts requiring detailed depiction and analysis of static information (Ellis 2008).

Task complexity was another important factor, with more complex tasks (resource-directing + resource-dispersing) being represented more in both within-group and between-group studies. Cognitively complex tasks are often designed to better capture the nuances of L2 learning and performance, providing richer data for analysis (Robinson 2005). Researchers might prefer such tasks to explore the effects of task sequencing more comprehensively (; Abdi Tabari and Cho 2022; Abdi Tabari and Miller 2021; Robinson 2010). The delivery method of tasks, whether face-to-face or computer-mediated, also showed variation. There were more face-to-face tasks in within-group studies and more computer-mediated tasks in between-group studies. Face-to-face tasks allow for better control and immediate feedback within groups, while computer-mediated tasks facilitate data collection and analysis across different groups, especially in remote or large-scale studies (Lai and Li 2011).

Regarding CALF constructs, the representation varied, with limited samples focusing on lexical complexity within within-group studies. Different constructs might be prioritized based on the study's theoretical framework or research questions (Vercellotti 2017). For instance, syntactic complexity and accuracy are often primary indicators of language proficiency and learning progress, whereas lexical complexity might be underreported due to the difficulty in quantifying it consistently (Crossley et al. 2015). Additionally, the coding of control variables like publication type and methodology indicated diverse research approaches and dissemination channels. The variation in publication types (journal articles, book chapters, dissertations, conference presentations) reflects the broad scope of dissemination and

academic inquiry in TS studies (Mackey and Gass 2016). The distinction between quasi-experimental and experimental methodologies highlights the different levels of control and randomization applied in the research designs (Plonsky 2017). Overall, the results for RQ1 reveal the methodological diversity and certain limitations inherent in the existing body of TS research in L2 learning. The identified methodological differences, represented by the moderator variables and their values, provide insight into the various approaches and contexts of these studies. While these differences pose challenges for data synthesis, they also underscore the richness and complexity of research in the TBLT field, highlighting the need for standardized reporting practices to enhance comparability and meta-analytical endeavors (Norouzian and Bui 2023).

The findings for RQ3 regarding the moderating effects of methodological differences in TS can be explained and justified by considering several factors related to the design and implementation of the studies. Firstly, the significant moderating effect of task complexity in both within-group and between-group analyses underscores the importance of cognitive engagement in language learning. More complex TS tasks, which involve both resource-directing and resource-dispersing elements, require learners to process language at a deeper cognitive level and integrate multiple linguistic components simultaneously. This aligns with the Cognition Hypothesis and the SSARC Model, which posit that learners need to be challenged within their cognitive capacities to maximize learning. By increasing task complexity, learners are encouraged to utilize higher-order thinking skills, leading to better retention and application of language skills (Robinson 2010). This increased cognitive demand can foster deeper linguistic processing, enhancing both explicit and implicit language knowledge (Malicka 2018).

Secondly, the role of L2 proficiency as a moderator, particularly in the within-group samples, suggests that learners' existing language skills influenced how they benefitted from TS. Learners with lower proficiency levels might struggle with complex tasks, reducing the effectiveness of TS for them, while those with higher proficiency might better handle and benefit from these tasks. This aligns with Ellis' (2017) assertion that task difficulty should match learners' current capabilities to maximize learning. The moderating effect of modality indicated that the medium through which tasks were delivered (written vs. oral) also impacted the effectiveness of TS. This may be due to the inherent differences in cognitive and linguistic demands between written and oral tasks. Written tasks often allow for more time to process and produce language, which can lead to more thoughtful and accurate responses, whereas oral tasks require real-time processing, which might be more challenging but also more beneficial for developing fluency (Kormos 2023).

The significant effect of the construct variable, especially in between-group studies, underscores the importance of targeting specific linguistic aspects such as accuracy. TS seemed to provide a clearer and more immediate impact on accuracy because this construct is a fundamental aspect of language proficiency that directly impacts communication effectiveness (Skehan et al. 2024). Constructs like syntactic and lexical complexity and fluency, while important, may require longer periods to show significant improvement and might be influenced by a broader range of external factors (Johnson 2017).

Including multiple moderator variables in the meta-regression models allowed for a more nuanced understanding of how these factors interact and influence the outcomes. The finding that task complexity and construct remained significant even when controlling for other variables suggests that these are robust predictors of TS effectiveness (Thompson 2014b). This highlights the need for instructional designs that incorporate varying levels of task complexity and focus on specific constructs to optimize L2 learning outcomes (Allaw and McDonough 2019). To recap, the moderating effects of task complexity, L2 proficiency, modality, and construct on the effectiveness of TS emphasize the importance of matching task demands to learners' abilities, the mode of task delivery, and the targeted linguistic outcomes. These factors should be carefully considered in both research and practical applications to maximize the benefits of TS in L2 learning.

7 Conclusion: implications, limitations, and future directions

This meta-analysis highlights the significant pedagogical implications of task sequencing (TS) in L2 learning. TS, especially when tasks are arranged from simple to complex, has been shown to improve learners' L2 performance, particularly when tasks increase in complexity over time. This suggests that TS is an effective tool for enhancing both oral and written language skills by offering learners ample opportunities to practice language in a variety of contexts, supporting their overall linguistic development.

The impact of TS is particularly evident in controlled settings where learners are exposed to tasks repeatedly. Instructors should aim to create supportive learning environments, such as small-group instruction or individualized feedback sessions, where similar tasks are practiced consistently. By tailoring tasks to learners' proficiency levels, TS can maximize cognitive engagement and help learners progress effectively. Given the diversity in learner characteristics, it is important to adapt TS strategies to fit the unique needs of each classroom.

The effectiveness of TS is also influenced by learners' proficiency levels, underscoring the importance of matching tasks to students' language abilities. Instructors should design task sequences that challenge learners without overwhelming them. For lower-proficiency learners,

simpler tasks may be more beneficial, while higher-proficiency learners can handle more complex tasks that stimulate higher-order cognitive skills. Properly matching task difficulty to learners' abilities ensures optimal cognitive load and promotes effective learning.

Task modality – whether written or oral – also plays a crucial role in L2 development. Written tasks allow more time for reflection and refinement, helping learners improve accuracy and depth in their responses. In contrast, oral tasks require real-time processing, fostering fluency and spontaneous language use. A balanced approach that includes both written and oral tasks in TS frameworks can address different aspects of language proficiency, improving both accuracy and fluency. For instance, a task sequence could begin with writing exercises focused on form and structure, then progress to oral tasks that require learners to apply their knowledge in dynamic, communicative settings.

In EFL contexts, where learners have limited exposure to the target language outside the classroom, TS can bridge the gap between instruction and real-world language use, helping students engage with the language more meaningfully. In ESL settings, where learners are immersed in the language, TS can further refine their skills through more advanced tasks that mirror real-life situations. Therefore, instructors in both EFL and ESL contexts should integrate TS into their teaching to meet the evolving needs of learners.

The study faced several limitations that merit acknowledgment, as they have implications for the interpretation and generalizability of the findings. A primary limitation was the uneven distribution of L2 proficiency levels among the participants. The absence of advanced-proficiency learners in within-group studies and the underrepresentation of low-proficiency learners in between-group studies may have skewed the results. This imbalance likely stemmed from challenges in recruiting advanced learners or the tendency to focus on intermediate learners, who are more commonly represented in language programs (Vuogan and Li 2023). Such uneven sampling restricts the study's ability to generalize findings across the full spectrum of L2 learners. Moreover, the predominance of EFL over ESL settings in the analyzed studies further limits the applicability of the results to more diverse learning environments, where learners face different linguistic and sociocultural challenges (Rose et al. 2021).

A second limitation arises from the study's approach to task sequencing. Rather than examining specific sequencing patterns – such as simple-medium-complex (SC), complex-medium-simple (CS), randomized (RAN), or no sequencing – it focused on synthesizing the overall effect sizes of task sequencing as a broader construct. While this approach offers valuable insights into the general effectiveness of task sequencing in L2 instruction, it falls short of providing granular evidence about the relative impact of different sequencing patterns.

Such evidence could inform targeted pedagogical strategies tailored to various learner needs. Future research should aim to disentangle these sequencing effects to refine instructional practices and offer more precise guidance for educators.

Adding to these challenges is the variation in task modalities and types across the studies reviewed. Within-group studies tended to employ more written tasks, whereas between-group studies focused primarily on oral tasks. This disparity complicates the comparability of results, as written and oral tasks impose distinct cognitive and linguistic demands on learners (Kormos 2023). Furthermore, differences in task types (e.g., narration versus description) and task complexity levels may have influenced the outcomes, introducing variability that limits the ability to draw consistent conclusions (Alexopoulou et al. 2017). These methodological inconsistencies were further exacerbated by variations in research design, including differences in how control variables were coded and the use of quasi-experimental versus experimental approaches. Such diversity in methodological rigor and dissemination channels underscores the complexity of synthesizing findings in task sequencing research (Plonsky 2017).

These limitations collectively highlight the need for future research to address the methodological gaps identified. First, achieving a more balanced representation of L2 proficiency levels across both within-group and between-group studies is essential. Including a wider range of proficiency levels, especially advanced learners, will improve the generalizability of findings. Second, researchers should strive to balance the representation of EFL and ESL contexts to better understand how task sequencing functions in different learning environments. Third, standardizing task modalities and types – ensuring a consistent mix of written and oral tasks and uniform task categories – will enhance the comparability of results across studies. Additionally, future research should explore the effects of task complexity in more controlled settings to provide deeper insights into the cognitive demands influencing L2 learning outcomes. Expanding the range of constructs measured, such as lexical complexity, will also contribute to a more comprehensive understanding of language proficiency development.

Finally, addressing these methodological limitations has broader implications for instructional practices. Consistent reporting of methodological details – such as clear coding of control variables and transparent distinctions between research designs – will facilitate more robust data synthesis and meta-analytical endeavors. By doing so, researchers can provide stronger evidence to support evidence-based instructional strategies.

In conclusion, this study sheds light on the critical role of methodological differences in shaping the effectiveness of task sequencing in L2 learning. The findings emphasize the importance of key moderating factors, including task complexity, L2 proficiency, modality, and targeted linguistic constructs. These insights underscore the need for carefully designed instructional interventions that align task difficulty with learners' abilities, consider the mode of task delivery, and address specific linguistic outcomes. Future research that builds on these findings will help educators create more effective and inclusive strategies for fostering L2 development.

Research ethics: The study was conducted following the Declaration of University of Nevada, Reno.

Informed consent: Not applicable.

Author contributions: The authors have accepted responsibility for the entire content of this manuscript and approved its submission.

Use of Large Language Models, AI and Machine Learning Tools: None declared.

Conflict of interest: All other authors state no conflict of interest.

Research funding: None declared.

Data availability: The raw data can be obtained on request from the corresponding author.

References

Abdi Tabari, Mahmoud & Michol Miller. 2021. Unraveling the effects of task sequencing on the syntactic complexity, accuracy, lexical complexity, and fluency of L2 written production. *Canadian Journal of Applied Linguistics* 24. 1–29.

Abdi Tabari, Mahmoud & Minyoung Cho. 2022. Task sequencing and L2 writing development: Exploring the SSARC model of pedagogic task sequencing. *Language Teaching Research*. https://doi.org/10.1177/13621688221090922.

Abdi Tabari, Mahmoud, Mark D. Johnson & Mahsa Farahanynia. 2023. Automated analysis of cohesive features in L2 writing: Examining effects of task complexity and task repetition. *Assessing Writing* 58. 100783.

Abdi Tabari, Mahmoud, Yizhou Wang & Michol Miller. 2024a. Understanding the SSARC model of task sequencing: Assessing L2 writing development. *Assessing Writing* 62. 100893.

Abdi Tabari, Mahmoud, Mark D. Johnson & Mahsa Farahanynia. 2024b. Task sequencing and L2 production: A research synthesis and meta-analysis. *International Journal of Applied Linguistics*. https://doi.org/10.1111/ijal.12670.

Alexopoulou, Theodora, Marije Michel, Akira Murakami & Detmar Meurers. 2017. Task effects on linguistic complexity and accuracy: A large-scale learner corpus analysis employing natural language processing techniques. *Language Learning* 67. 180–208.

Allaw, Elissa. 2017. *The effect of the SSARC model of task sequencing on L2 written production*. Canada: Concordia University Dissertation.

Allaw, Elissa & Kim McDonough. 2019. The effect of task sequencing on second language written lexical complexity, accuracy, and fluency. *System* 85. 102014.

Awwad, Anas & Parvaneh Tavakoli. 2019. Task complexity, language proficiency, and working memory: Interaction effects on second language speech performance. *International Review of Applied Linguistics in Language Teaching* 60. 169–196.

Baralt, Melissa. 2014. Task complexity and task sequencing in traditional versus online language classes. In Melissa Baralt, Roger Gilabert & Peter Robinson (eds.), *Task sequencing and instructed second language learning*, 95–122. London: Bloomsbury.

Baralt, Melissa, Roger Gilabert & Peter Robinson. 2014. An introduction to theory and research in task sequencing and instructed second language learning. In Melissa Baralt, Roger Gilabert & Peter Robinson (eds.), *Task sequencing and instructed second language learning*, 1–37. London: Bloomsbury.

Barrot, Jessie & Mari K. Gabinete. 2019. Complexity, accuracy, and fluency in the argumentative writing of ESL and EFL learners. *International Review of Applied Linguistics in Language Teaching* 59. 209–232.

Bryfonski, Lara & Todd H. McKay. 2017. TBLT implementation and evaluation: A meta-analysis. *Language Teaching Research* 23(5). 603–632.

Cohen, Jacob. 1988. *Statistical power analysis for the behavioral sciences*, 2nd edn. Hillsdale, NJ: Lawrence Erlbaum Associates.

Crossley, Scot A., Tom Salsbury & Danielle S. Mcnamara. 2015. Assessing lexical proficiency using analytic ratings: A case for collocation accuracy. *Applied Linguistics* 36. 570–590.

Ellis, Rod. 2008. *The study of second language acquisition*. Oxford: Oxford University Press.

Ellis, Rod. 2017. Position paper: Moving task-based language teaching forward. *Language Teaching* 50(4). 507–526.

Ellis, Rod, Peter Skehan, Shaofeng Li, Natsuko Shintani & Craig Lambert. 2020. *Task-based language teaching: Theory and practice*. Cambridge: Cambridge University Press.

Farahanynia, Mahsa. 2019. The impact of task complexity and task sequencing on L2 written performance: Examining the SSARC model of task sequencing. In *The 17th International Teaching English Language and Literature Society of Iran Conference*. Tabriz, Iran: Tabriz Islamic University.

Faruji, Laleh F. & Farid Ghaemi. 2017. Task complexity manipulation and accuracy in writing performance. *Journal of Teaching Language Skills* 35. 103–132.

Granfeldt, Jonas. 2007. Speaking and writing in L2 French: Exploring effects on fluency, complexity, and accuracy. In Siska van Daele, Alex Housen, Folkert Kuiken, Michel Pierrard & Inneke Vedder (eds.), *Complexity, accuracy and fluency in second language use, learning & teaching*, 87–98. Brussels: University of Brussels.

Hedges, Larry V. 2019. The statistics of replication. *Methodology* 15. 3–14.

Jackson, Daniel O. & Sakol Suethanapornkul. 2013. The cognition hypothesis: A synthesis and meta-analysis of research on second language task complexity. *Language Learning* 63. 330–367.

Jin, Chaoyang & Jing Yan. 2024. The effects of task complexity and task sequencing on L2 performance: A systematic review. *The Language Learning Journal* 53(1). 1–28.

Johnson, Mark D. 2017. Cognitive task complexity and L2 written syntactic complexity, lexical complexity, accuracy, and fluency: A research synthesis and meta-analysis. *Journal of Second Language Writing* 37. 13–38.

Johnson, Mark D. & Mahmoud Abdi Tabari. 2022. Task planning and oral L2 production: A research synthesis and meta-analysis. *Applied Linguistics* 43. 1143–1164.

Johnson, Mark D. & Mahmoud Abdi Tabari. 2023. Planning in L2 writing: A research synthesis and meta-analysis. *System* 118. 103152.

Kellogg, Ronald T. 1996. A model of working memory in writing. In C. Michael Levy & Sara Ransdell (eds.), *The science of writing: Theories, methods, individual differences, and application*, 57–71. Mahwah, NJ: Lawrence Erlbaum Associates.

Kim, Nayoung. 2023. The effects of different task sequences on novice L2 learners' oral performance in the classroom. *Language Teaching Research* 27. 415–440.

Kim, Hyunwoo & Haerim Hwang. 2022. Assessing verb-construction integration in young learners of English as a foreign language: Analyses of written and spoken production. *Language Learning* 72. 497–533.

Kormos, Judit. 2023. The role of cognitive factors in second language writing and writing to learn a second language. *Studies in Second Language Acquisition* 45. 622–646.

Korvesi, Evgenia & Marije Michel. 2022. Chatting with your peers across modalities: Effects of performing increasingly complex written computer-mediated tasks on oral L2 development. *Languages* 7. 276.

Lai, Chun & Guofang Li. 2011. Technology and task-based language teaching: A critical review. *CALICO Journal* 28. 498–521.

Lee, Jiyong. 2021. Task sequencing and task complexity effects on L2 Writing: Does task order really matter? *The Journal of Asia TEFL* 18. 1088–1107.

Levelt, Willem. 1989. *Speaking: From intention to articulation*. Cambridge: Cambridge University Press.

Levkina, Mayya. 2013. *The role of task sequencing in L2 development as mediated by working memory capacity*. Spain: Universität de Barcelona Dissertation.

Levkina, Mayya & Roger Gilabert. 2014. Task sequencing in the L2 development of spatial expressions. In Melissa Baralt, Roger Gilabert & Peter Robinson (eds.), *Task sequencing and instructed second language learning*, 37–70. London: Bloomsbury.

Mackey, Alison & Susan M. Gass. 2016. *Second language research: Methodology and design*, 2nd edn. New York: Routledge.

Malicka, Aleksandra. 2014a. *The role of task and task sequencing in L2 monologic oral performance*. Spain: Universität de Barcelona Dissertation.

Malicka, Aleksandra. 2014b. The role of task sequencing in L2 oral production. In Melissa Baralt, Roger Gilabert & Peter Robinson (eds.), *Task sequencing and instructed second language learning*, 71–95. London: Bloomsbury.

Malicka, Aleksandra. 2018. The role of task sequencing in fluency, accuracy, and complexity: Investigating the SSARC model of pedagogic task sequencing. *Language Teaching Research* 24(5). 1–24.

Malicka, Aleksandra & Shoko Sasayama. 2017. The importance of learning from the accumulated knowledge: Findings from a research synthesis on task complexity. In *The 7th Biennial International Conference on Task-Based Language Teaching*. Spain: Universität de Barcelona.

Norouzian, Reza & Gavin Bui. 2023. Meta-analysis of second language research with complex research designs. *Studies in Second Language Acquisition* 46(1). 1–25.

Ortega, Lourdes. 2009. Studying writing across EFL contexts: Looking back and moving forward. In Rosa Manchón (ed.), *Writing in foreign language contexts: Learning, teaching, and research*, 209–231. Multilingual Matters.

Page, Matthew J., Joanne E. McKenzie, Patrick M. Bossuyt, Isabelle Boutron, Tammy C. Hoffmann, Cynthia D. Mulrow, David Moher, Jennifer M. Tetzlaff, Elie A. Akl, Sue E. Brennan, Roger Chou, Julie Glanville, Jeremy M. Grimshaw, Asbjørn Hróbjartsson, Manoj M. Lalu, Tianjing Li, Elizabeth W. Loder, Evan Mayo-Wilson, Steve McDonald, Luke A. McGuinness, Lesley A. Stewart, James Thomas, Andrea C. Tricco, Vivian A. Welch & Penny Whiting. 2021. The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *BMJ* 372. S312–S313.

Pang, Francine & Peter Skehan. 2014. Self-reported planning behaviour and second language performance in narrative retelling. In Peter Skehan (ed.), *Processing perspectives on task performance*, 95–128. London: John Benjamins.

Plonsky, Luke. 2017. Quantitative research methods in instructed SLA. In Shawn Loewen & Masatoshi Sato (eds.), *The Routledge handbook of instructed second language acquisition*, 505–521. London: Routledge.

Plonsky, Luke & You J. Kim. 2016. Task-based learner production: A substantive and methodological review. *Annual Review of Applied Linguistics* 36. 73–97.

Plonsky, Luke & Frederick L. Oswald. 2014. How big is ‘big’? Interpreting effect sizes in L2 research. *Language Learning* 64. 878–912.

Robinson, Peter. 2001. Task complexity, task difficulty, and task production: Exploring interactions in a componential framework. *Applied Linguistics* 22(1). 27–57.

Robinson, Peter. 2005. Cognitive complexity and task sequencing: Studies in a componential framework for second language task design. *International Review of Applied Linguistics in Language Teaching* 43. 1–32.

Robinson, Peter. 2010. Situating and distributing cognition across task demands: The SSARC model of pedagogic task sequencing. In Martin Pütz & Laura Sicola (eds.), *Cognitive processing in second language acquisition: Inside the learner’s mind*, 243–268. Amsterdam: John Benjamins.

Rose, Heath, Jim McKinley & Nicola Galloway. 2021. Global Englishes and language teaching: A review of pedagogical research. *Language Teaching* 54(2). 157–189.

Skehan, Peter. 1998. *A cognitive approach to language learning*. Oxford: Oxford University Press.

Skehan, Peter. 2016. Tasks vs. conditions: Two perspectives on task research and its implications for pedagogy. *Annual Review of Applied Linguistics* 36. 34–49.

Skehan, Peter, Gavin Bui, Zhan Wang & Sabrina Shum. 2024. Re-examining accuracy measures in second language task-based spoken performance. *Research Methods in Applied Linguistics* 3(1). 100098.

StataCorp. 2020. *Stata statistical software: Release 16.1*. College Station, TX: StataCorp LLC.

Thompson, Colin. 2014a. *Guided planning, task complexity, and second language oral development*. England: University of Central Lancashire Dissertation.

Thompson, Colin. 2014b. The effects of guided planning, task complexity, and task sequencing on L2 oral production. In Melissa Baralt, Roger Gilabert & Peter Robinson (eds.), *Task sequencing and instructed second language learning*, 123–148. London: Bloomsbury.

Tomasello, Michael. 2003. *Constructing a language: A usage-based theory of language acquisition*. Cambridge: Harvard University Press.

Vercellotti, Mary L. 2017. The development of complexity, accuracy, and fluency in second language performance: A longitudinal study. *Applied Linguistics* 38(1). 90–111.

Viechtbauer, Wolfgang. 2010. Conducting meta-analyses in R with the metafor package. *Journal of Statistical Software* 36. 1–48.

Vuogan, A. & S. Li. 2023. A systematic review of meta-analyses in second language research: Current practices, issues, and recommendations. *Applied Linguistics Review* 15(4). 1621–1644.

Weissgerber, Sophia C., Matthias Brunmair & Ralf Rummer. 2021. Null and void? Errors in meta-analysis on perceptual disfluency and recommendations to improve meta-analytical reproducibility. *Educational Psychology Review* 33(3). 1221–1247.

Yoo, Hyunbin & Hyunwoo Kim. 2023. A cross-modal analysis of lexical sophistication: EFL and ESL learners in written and spoken production. *International Review of Applied Linguistics in Language Teaching* 0(0). https://doi.org/10.1515/iral-2023-0204.

Supplementary Material: This article contains supplementary material (https://doi.org/10.1515/iral-2024-0219).

Bionotes

Mahmoud Abdi Tabari

Department of English, University of Nevada, Frandsen Humanities, Room 105, Reno, NV 89557, USA

Mabditabari@unr.edu

https://orcid.org/0000-0002-8022-5415

Mahmoud Abdi Tabari is a Teaching Assistant Professor of TESOL and Writing Studies at the University of Nevada, Reno (UNR). His research focuses on second language acquisition, second language writing, and task-based language teaching. He has published his work in various top-tier academic journals, including *Applied Linguistics, The Modern Language Journal, Journal of Second Language Writing, Language Teaching Research, System, Reading and Writing, Assessing Writing, and Journal of English for Academic Purposes*.

Hansol Lee
Department of English, Korea Military Academy, Seoul, Korea
https://orcid.org/0000-0002-6912-7128

Hansol Lee is a Professor of English at Korea Military Academy, South Korea. He received his PhD degree in Education from the University of California, Irvine. His research interests include methods and statistics, education, and literacy. His work has been published in *Review of Educational Research, Education Research Review, Child Development, Applied Measurement in Education, Applied Linguistics, Language Learning, The Modern Language Journal, Language Learning & Technology, Journal of Language and Social Psychology, Annual Review of Applied Linguistics*, among others.

Mahsa Farahanynia
Department of English Language, Pishtazan Higher Education Institute, Shiraz, Iran
https://orcid.org/0000-0003-3038-1871

Mahsa Farahanynia is an Assistant Professor at Pishtazan Higher Education Institute, Iran. Her research interests are centered on task-based language teaching, language assessment, and discourse analysis. She has published articles in a number of well-established journals, like *Educational Assessment, The Modern Language Journal, Assessing Writing, and System*.